Guided LAB 332.3.2 - Explore and Practice for Power Query Editor

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Lab Overview:

This lab introduces learners to the **Power Query Editor** in Excel, a tool designed for transforming and preparing data efficiently. Through a series of practical examples and hands-on exercises, learners will explore various Power Query features such as importing data, cleaning datasets, transforming columns, applying calculations, and handling missing values. These tasks are fundamental in preparing data for analysis, helping learners build real-world data transformation skills.

Sample Dataset:

➔ **Click here to Download Sample file.**

Learning Objective:

By the end of this lab, learners will be able to.

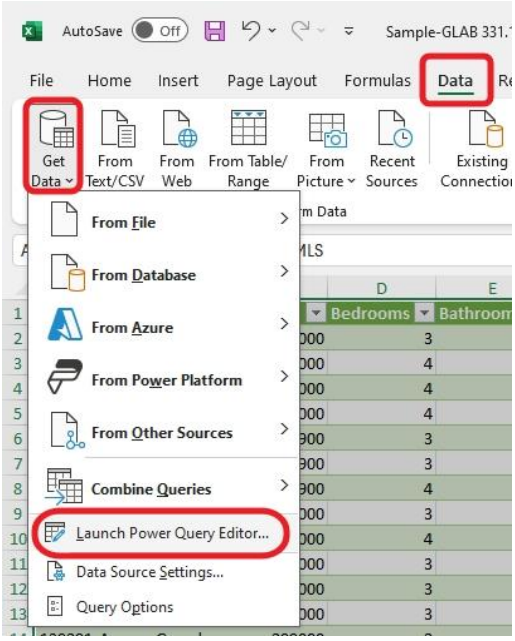
- How to import Data from CSV into Excel using Power Query
- Utilizing the Power Query Editor feature in Excel.
- Selecting and importing specific data tables.
- Cleaning and refreshing imported data.
- Handling data transformation and loading.
- Use Power Query's **Group By**, **Conditional Columns**, and other features for data aggregation and enhancement.
- Address common data issues such as missing values, duplicate records, and inconsistent formatting.
- Perform date and time manipulations like calculating age, extracting years, and splitting dates.

Overview Power Query Editor in Excel

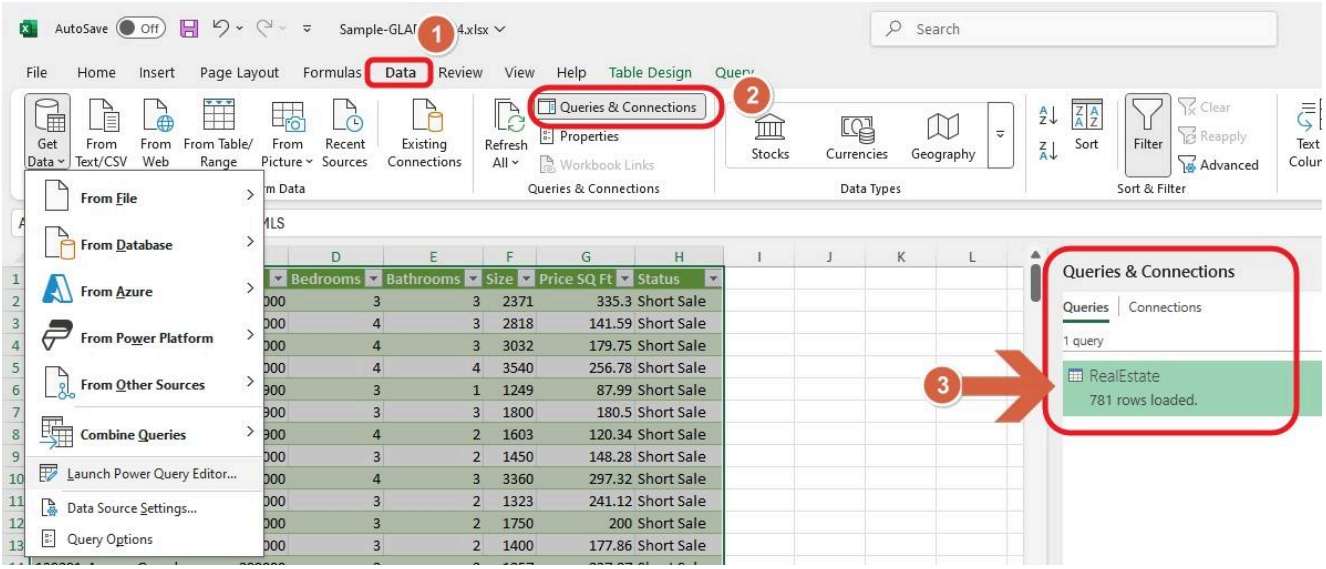
The Power Query Editor is the central component of all data transformations. It is a separate graphical interface within Excel that plays host to your queries. It is the primary workspace where you'll spend most of your time refining and reshaping your data.

You can access the editor in several ways:

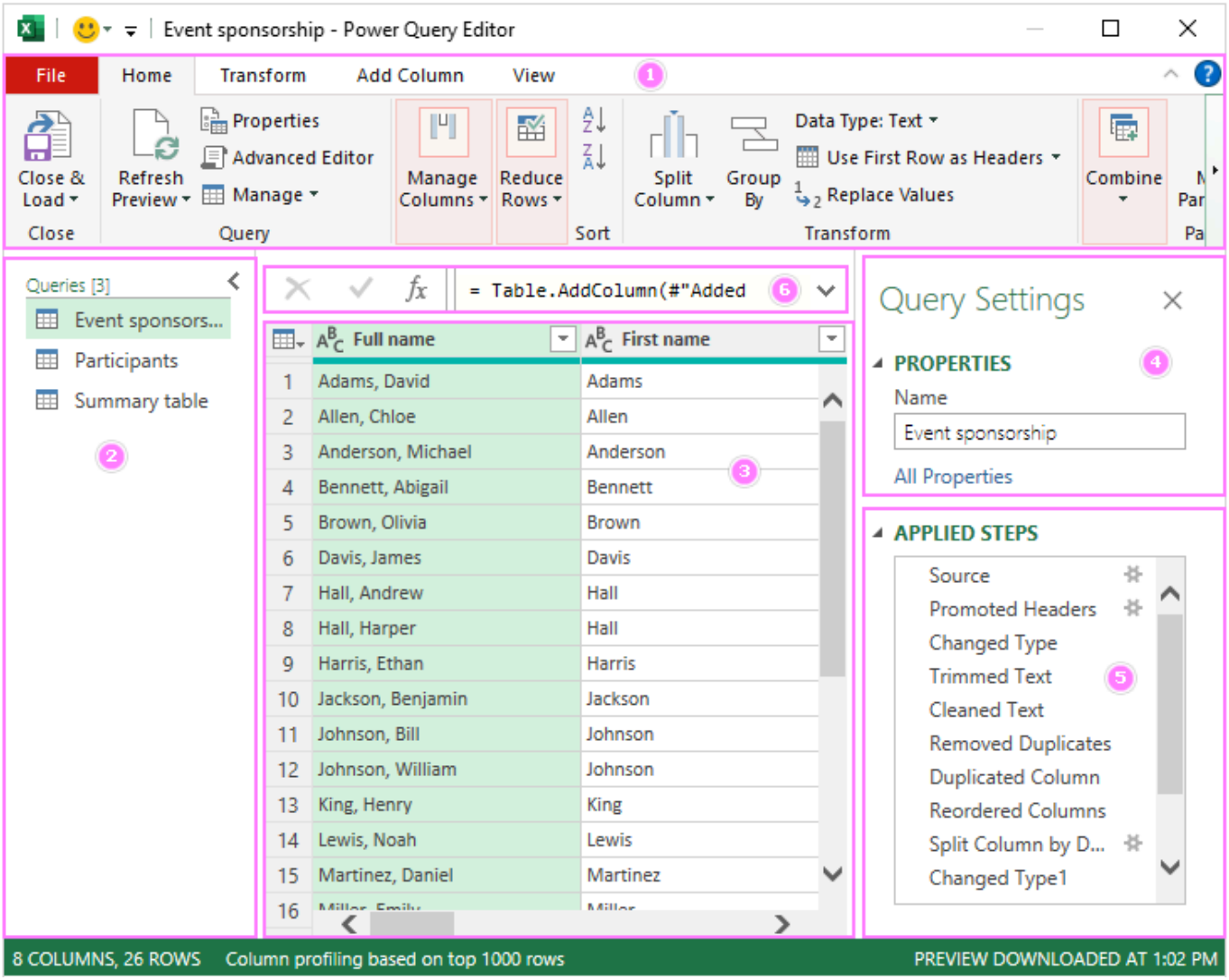
- **Option 1:** When you connect to a data source or create a new query, the Editor opens automatically.
- **Option 2:** On the **Data** tab, in the **Get & Transform** group, click **Get Data** and select **Launch Power Query Editor.**



- **Option 3:** Alternatively, on the **Data** tab, click the **Queries & Connections** button to open this pane. Then, double-click a query in the pane to launch the editor, as shown in the screenshot below.

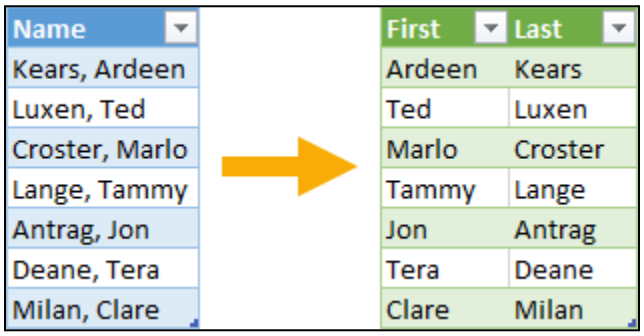

AutoSave Off
Sample-GLAI 4.xlsx
Search
File Home Insert Page Layout Formulas Data Review View Help Table Design Query
Get Data
From Text/CSV
From Web
From Table/ Range
From Picture
Recent Sources
Existing Connections
Refresh All
Queries & Connections
Properties
Workbook Links
Stocks
Currencies
Geography
Sort
Filter
Clear
Reapply
Advanced
Text Colu
From File
From Database
From Azure
From Power Platform
From Other Sources
Combine Queries
Launch Power Query Editor...
Data Source Settings...
Query Options
Queries & Connections
Data Types
Sort & Filter
Bedrooms Bathrooms Size Price SQ Ft Status
Queries & Connections
Queries Connections
1 query
RealEstate
781 rows loaded.

Power Query Editor



Begin:

Example 1. Split cells by delimiter



Name
Kears, Ardeen
Luxen, Ted
Croster, Marlo
Lange, Tammy
Antrag, Jon
Deane, Tera
Milan, Clare

First	Last
Ardeen	Kears
Ted	Luxen
Marlo	Croster
Tammy	Lange
Jon	Antrag
Tera	Deane
Clare	Milan

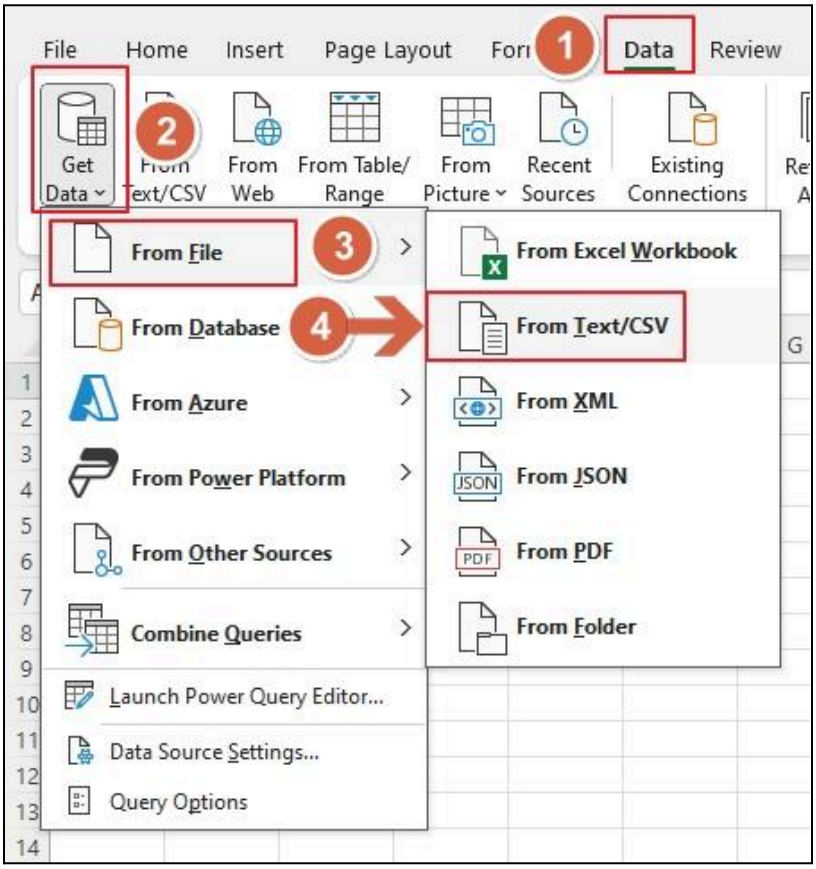
Here's the scenario:

- The data contains a list of names and they're all formatted the same way. Last name followed by a comma and space and then the first name.
- But you need the first and last names separated in their columns. You can do this with Power Query's split column command.

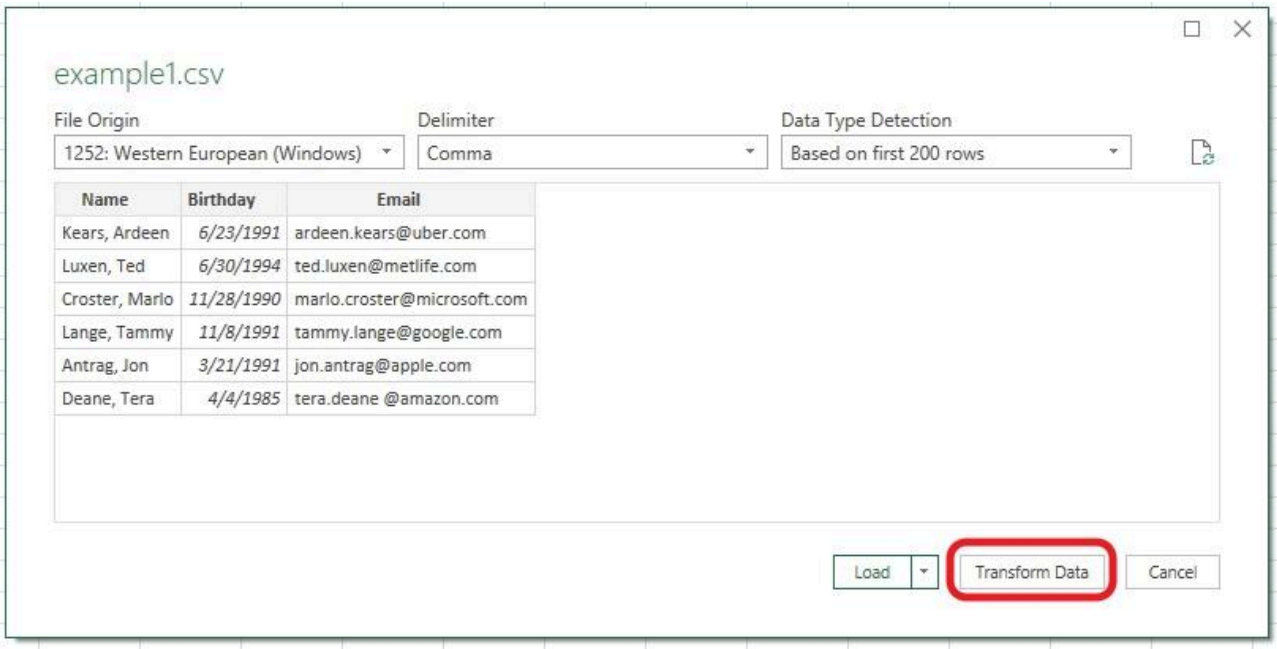
In this example, we will use `**example1.csv**` file

Follow the following steps:

1. Open Excel
2. Click New, then click on Blank workbook
3. Go to **Data** tab → **Get Data** → **From File** → click on **From Text/CSV**, as shown in the screenshot below.



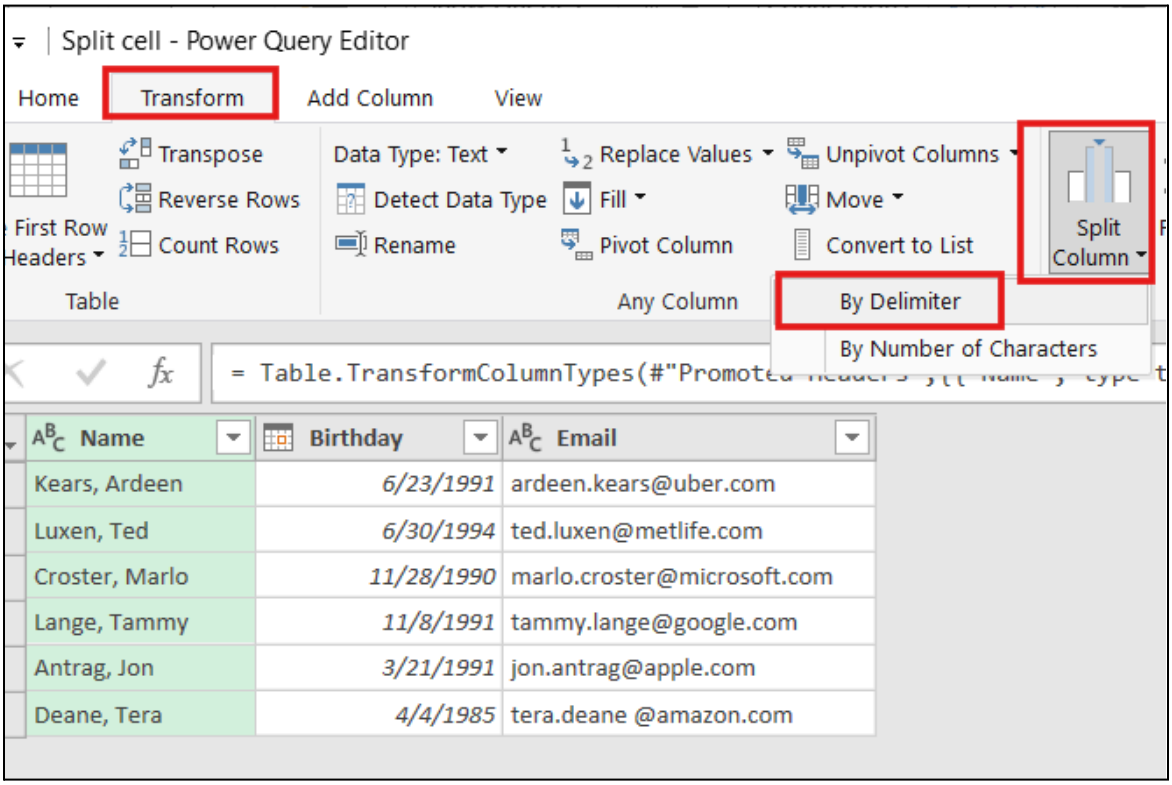
4. In the ***Import** Data* dialog box, select the desired CSV file to import it; in our case, select **example.csv**. click **Import**. As shown in the screenshot below.
5. Click on the **Transform Data** button, as shown below.



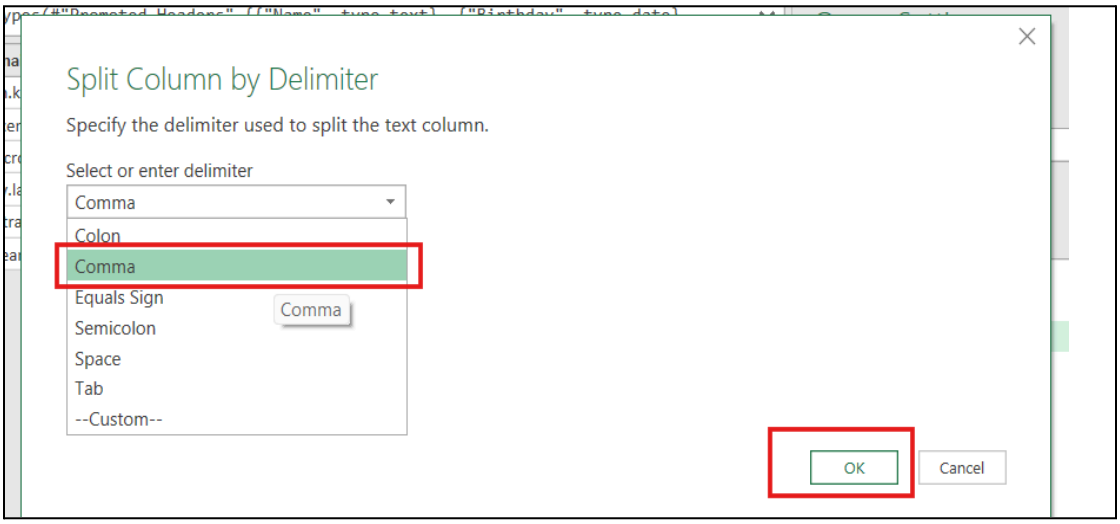
Clicking on the `**Transform Data**` button will bring you to a **Power Query Editor**. This editor allows you to transform your data before loading it into a new worksheet.

6. In the query editor, select the column you want to split, in our case we will Select the Name column and click on the **Transform** tab → **Split Column** option → choose **By**

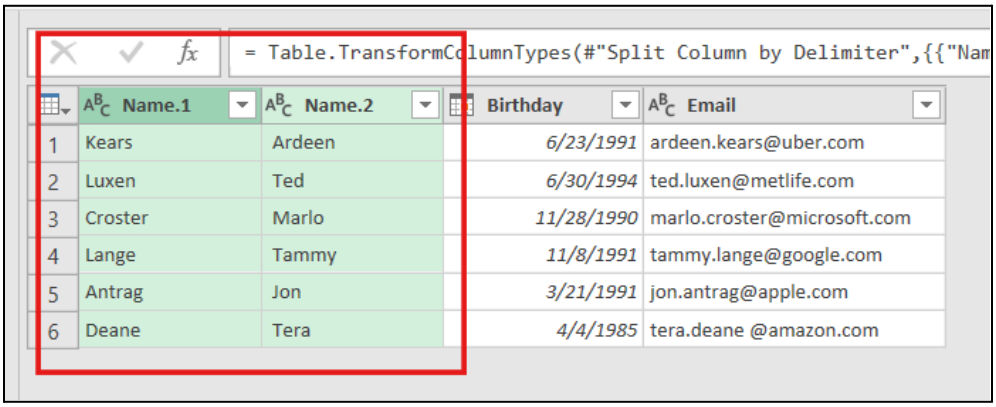
Delimiter. As shown in the screenshot below.



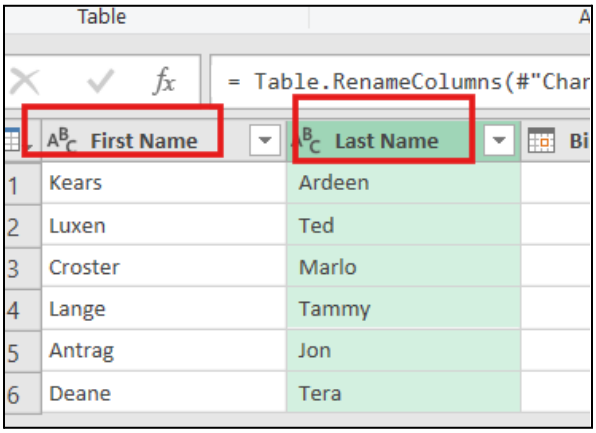
7. The **Split a column by delimiter** dialog box appears.
    - Here you can select delimiters such as **colon**, **comma**, **equals sign**, **Semicolon**, **space**, **tab**, or **custom.** You can also select **Custom** to specify any character delimiter.
8. In the **Select or enter a delimiter** drop-down, select **Comma,** and click on **OK** button. You can also select **Custom** to specify any character delimiter. As shown in the screenshot below.



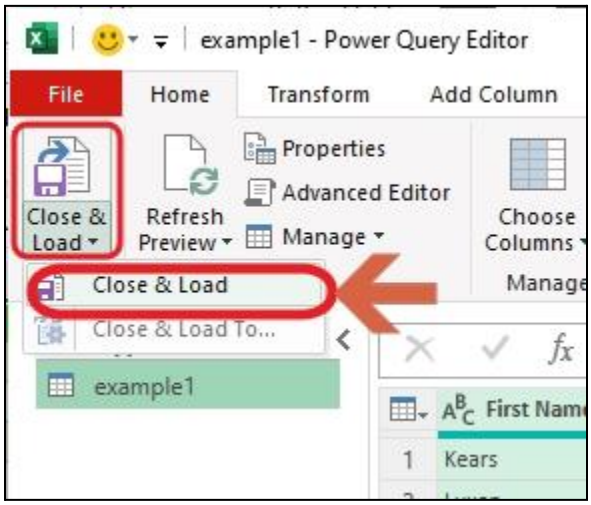
Now the names are all separated into their own columns. As shown in the screenshot below.



You can edit the field names by double-clicking the column name, such as **First Name** and **Last Name.**



9. We are now ready to load it into our Excel worksheet by selecting **Close & Load**



The final view of the data is shown in the screenshot below.

	First Name	Last Name	Birthday	Email
2	Kears	Ardeen	6/23/1991	ardeen.kears@uber.com
3	Luxen	Ted	6/30/1994	ted.luxen@metlife.com
4	Croster	Marlo	11/28/1990	marlo.croster@microsoft.com
5	Lange	Tammy	11/8/1991	tammy.lange@google.com
6	Antrag	Jon	3/21/1991	jon.antrag@apple.com
7	Deane	Tera	4/4/1985	tera.deane @amazon.com

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Name.1	Name.2	Birthday	Email
Kears	Ardeen	6/23/1991	ardeen.kears@uber.com
Luxen	Ted	6/30/1994	ted.luxen@metlife.com
Croster	Marlo	11/28/1990	marlo.croster@microsoft.com
Lange	Tammy	11/8/1991	tammy.lange@google.com
Antrag	Jon	3/21/1991	jon.antrag@apple.com
Deane	Tera	4/4/1985	tera.deane @amazon.com
***Split columns by delimiter (comma)**			

➔ [Click here to visit official website for information]

Example 2. Fix country-specific formatting

Product	Sales
Product A	10.346,36
Product B	9.731,03
Product C	43,76
Product D	45.987,78
Product E	716,35

Product	Sales
Product A	$10,346.36
Product B	$9,731.03
Product C	$43.76
Product D	$45,987.78
Product E	$716.35

Here is the scenario:

Different countries use different formatting conventions for things like dates and numbers.

- In Germany and many other mainland European countries, $10,346.36 would be written as $10.346,36. In this number formatting, the comma and period play the opposite roles, and Excel will treat it as text instead of a number.

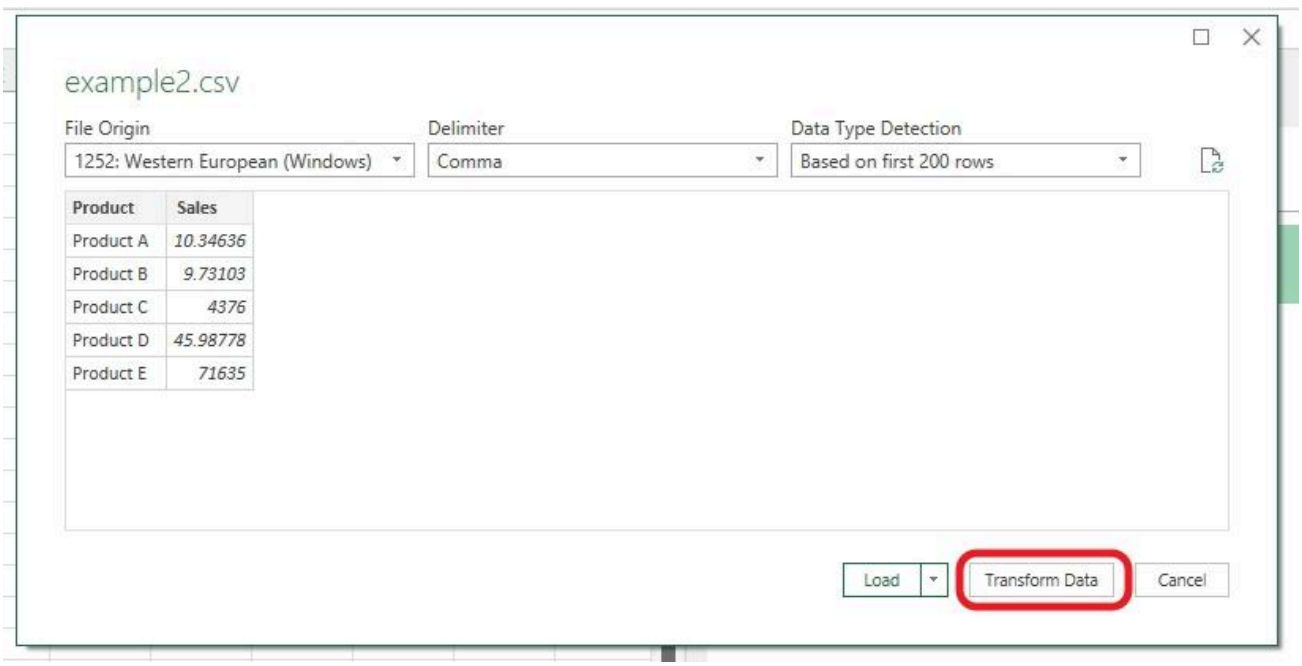
- Dates can be another tricky situation. In the US, they use an mm/dd/yyyy format, while the rest of the world uses the **dd/mm/yyyy** or **yyyy-mm-dd** format.

- The above issue can be time-consuming when trying to work with data received from different locations across the world. Power Query's locale functionality can make this a breeze to fix.

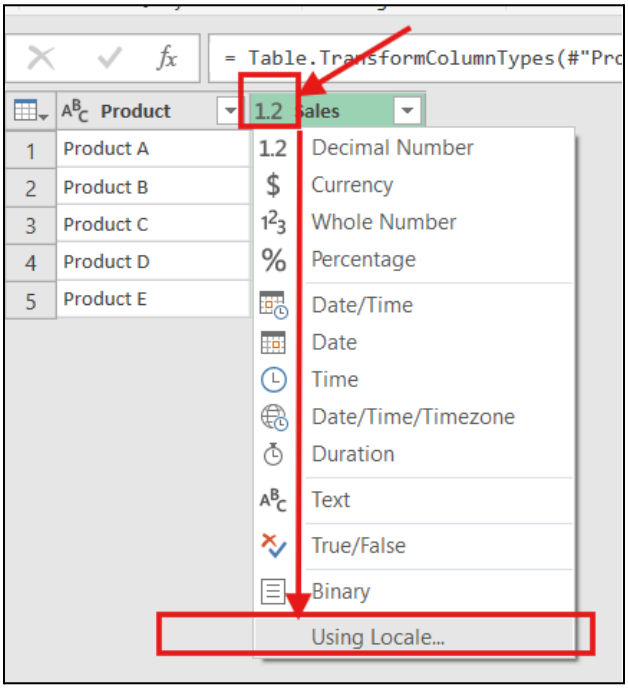
Follow the following steps:

In this example, we will use the **`example2.csv`** file for demonstration.

1. Open Excel
2. Click New, then click on Blank workbook
3. Go to **Data** tab → **Get Data** → **From File** → click on **From Text/CSV**.
4. In the **Import Data** dialog box, select the desired **CSV** file to import it; in our case, select **example2.csv**. Click the **Import** button.
5. Click on the **Transform Data** button, as shown below.

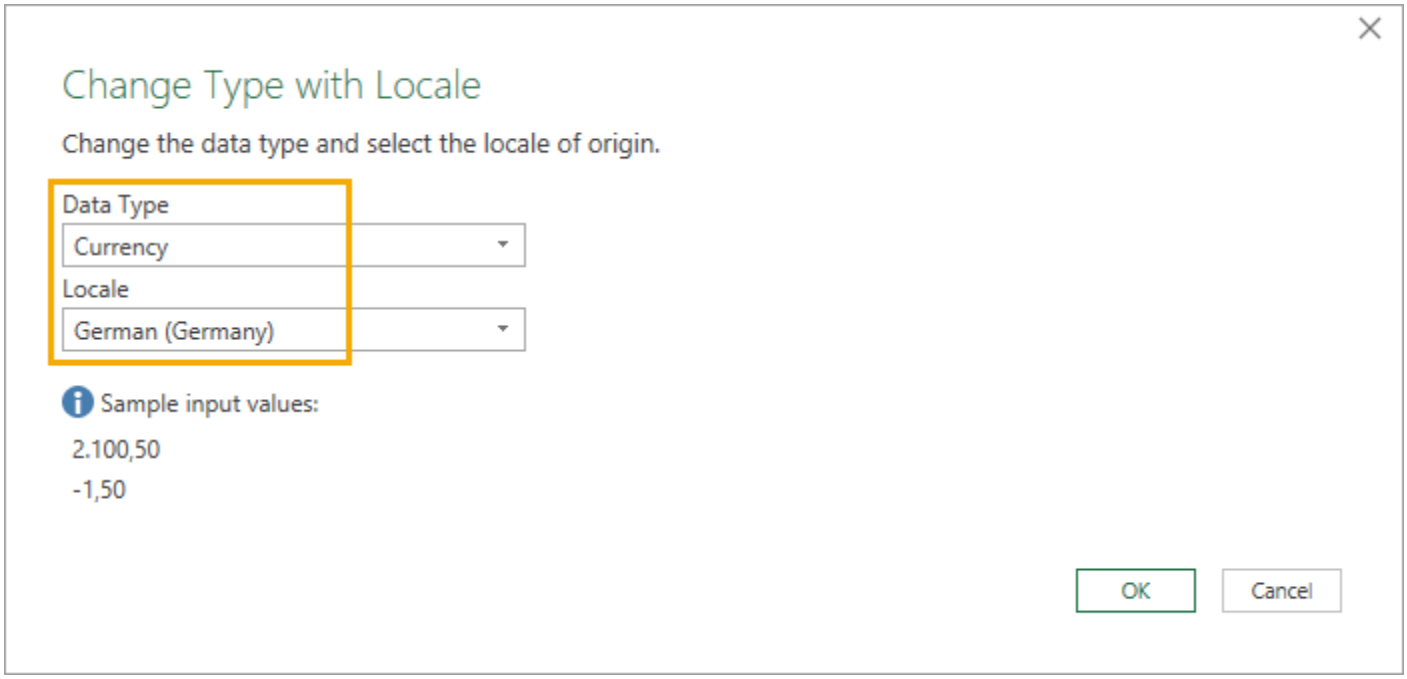


6. Left-click on the **Sales** column; the drop-down menu will open; choose `Using Locale` from the menu.

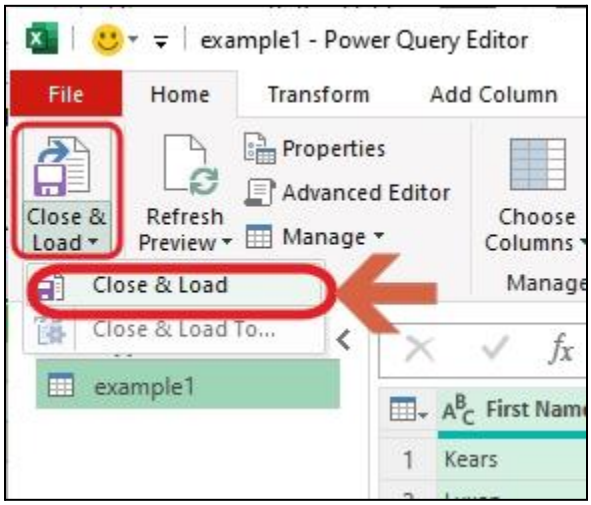


7. The `**Change Type with locale**` dialog box appears. We can select the Data Type that you're trying to transform. In this case,
    a. In Data type: Select **Currency**
    b. In Locale: Select **German (Germany)**
    c. Press the **OK** button

Now Excel will recognize the data as numbers instead of text!



8. We are now ready to load it into our Excel worksheet by selecting **Close & Load → Close & load, as shown in the screenshot below.**



The final view of the data is shown in the screenshot below.

Product	Sales
Product A	$10,346.36
Product B	$9,731.03
Product C	$43.76
Product D	$45,987.78
Product E	$716.35

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Product	Sales
Product A	10.3464
Product B	9.731
Product C	4376
Product D	45.9878
Product E	71635

Changed currency according to country (Germany) otherwise, if changed manually the values will be treated as string

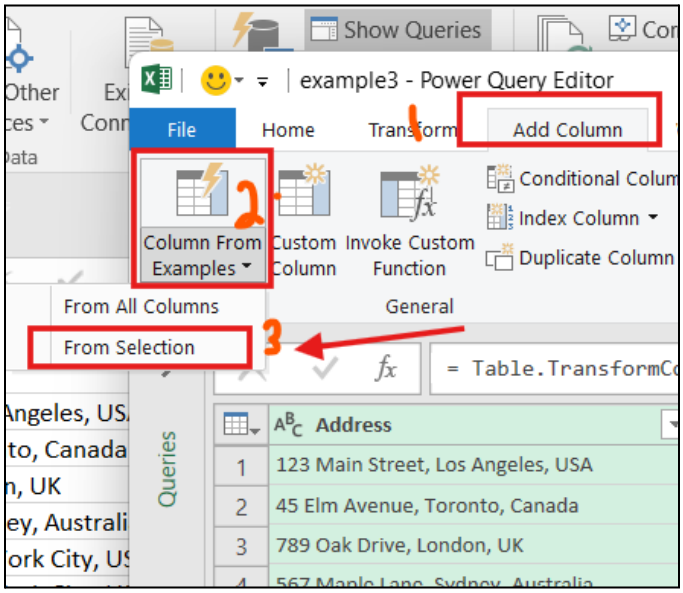
Example 3. Add `Column from Example`

In this example, we will explore the `Add a `Column From Examples` feature. You can use Power Query to extract country names from addresses in these situations.

In this example, we will use the **`example3.csv`** file for demonstration.

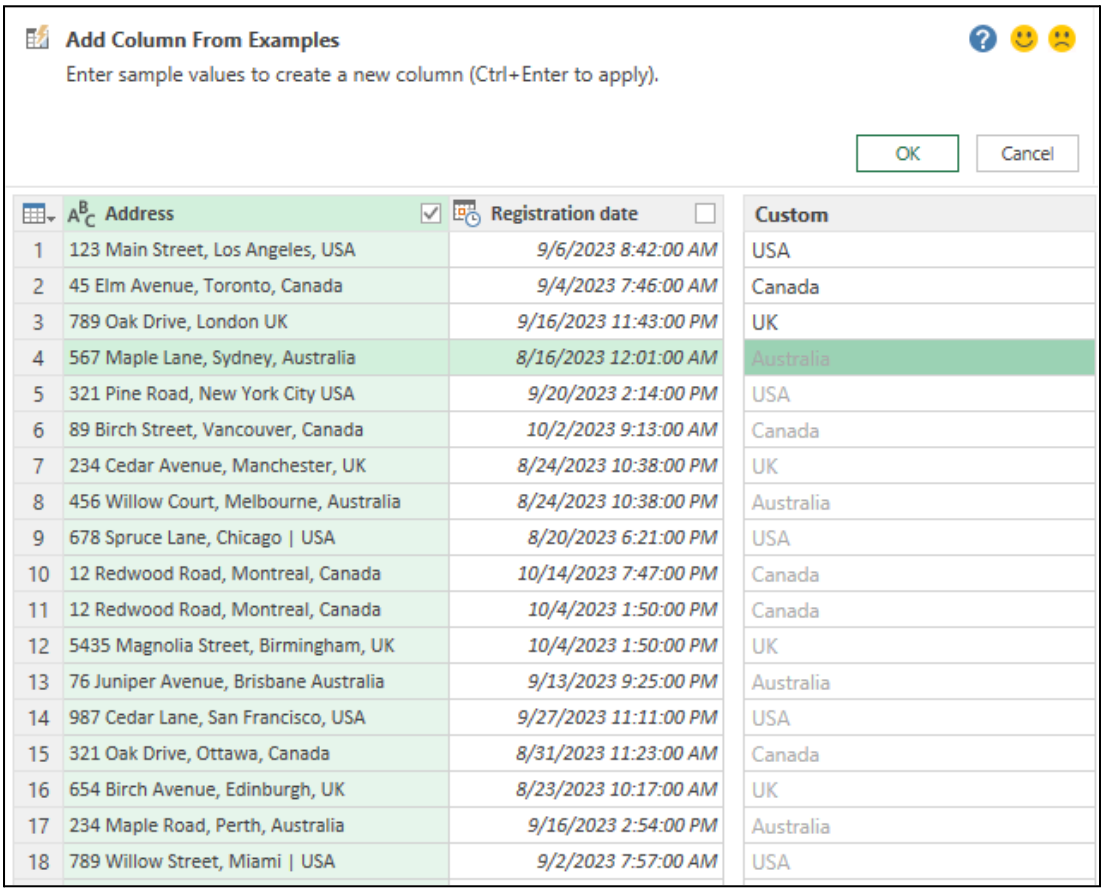
9. Open Excel
10. Click New, then click on Blank workbook
11. Go to **Data** tab → **Get Data** → **From File** → click on **From Text/CSV**.

12. In the *Import Data* dialog box, select the desired **CSV** file to import it; in our case, select **example3.csv**. Click the **Import** button.
13. Click on the **Transform Data** button.
14. Select the column that contains the source data (the Address column in our example).
15. On the ***Add Column*** tab, click `***Column from Examples***` → ***From Selection*, as shown in the screenshot below.**



16. In the first row of the new column, type the country name that corresponds to the first address. Power Query will try to infer the pattern and fill in the rest of the values based on your example. If some cells are blank or filled with incorrect values, provide another example in the second row or any other row until Power Query gets it right.

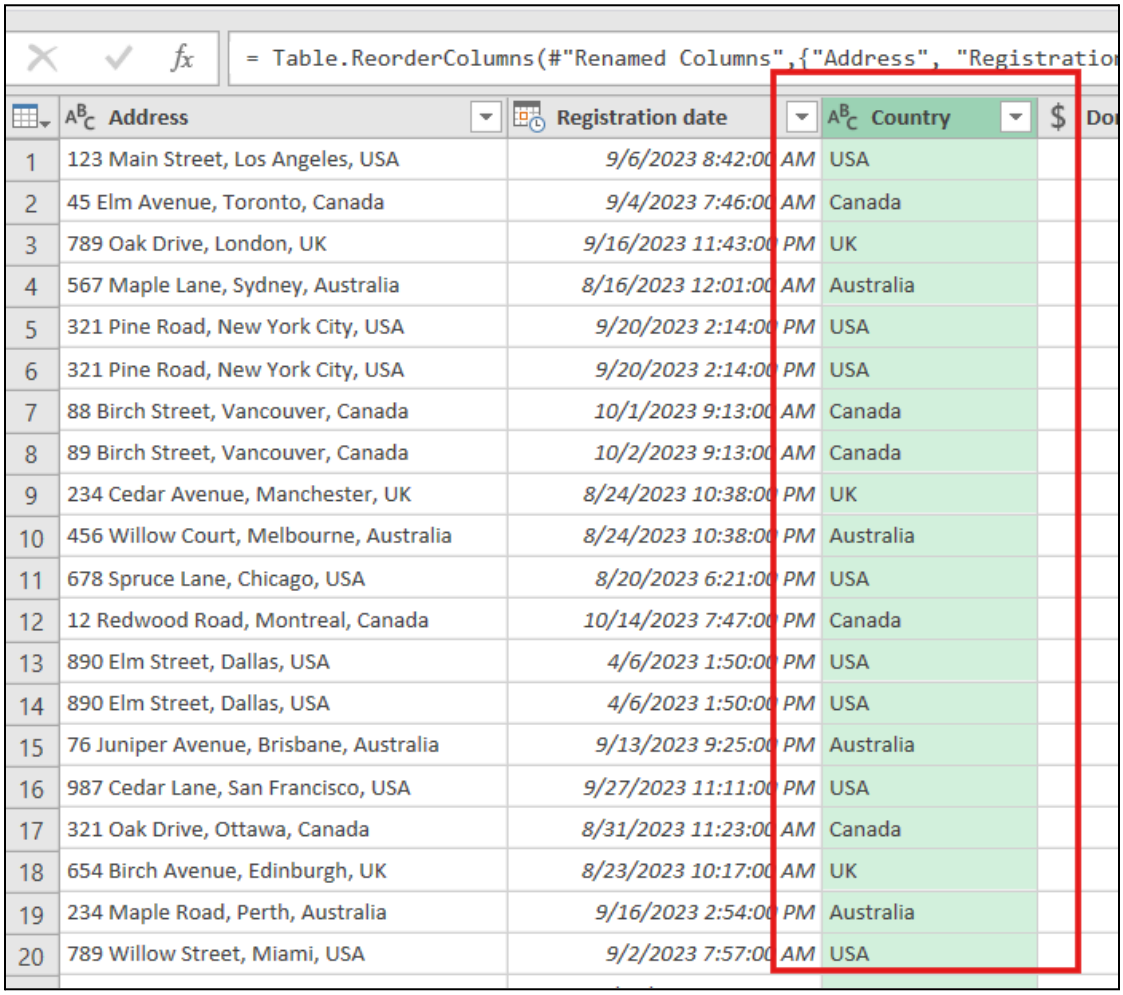
17. When all the cells are filled with the correct values, press Ctrl + Enter to apply the changes.



	Address	Registration date	Custom
1	123 Main Street, Los Angeles, USA	9/6/2023 8:42:00 AM	USA
2	45 Elm Avenue, Toronto, Canada	9/4/2023 7:46:00 AM	Canada
3	789 Oak Drive, London UK	9/16/2023 11:43:00 PM	UK
4	567 Maple Lane, Sydney, Australia	8/16/2023 12:01:00 AM	Australia
5	321 Pine Road, New York City USA	9/20/2023 2:14:00 PM	USA
6	89 Birch Street, Vancouver, Canada	10/2/2023 9:13:00 AM	Canada
7	234 Cedar Avenue, Manchester, UK	8/24/2023 10:38:00 PM	UK
8	456 Willow Court, Melbourne, Australia	8/24/2023 10:38:00 PM	Australia
9	678 Spruce Lane, Chicago | USA	8/20/2023 6:21:00 PM	USA
10	12 Redwood Road, Montreal, Canada	10/14/2023 7:47:00 PM	Canada
11	12 Redwood Road, Montreal, Canada	10/4/2023 1:50:00 PM	Canada
12	5435 Magnolia Street, Birmingham, UK	10/4/2023 1:50:00 PM	UK
13	76 Juniper Avenue, Brisbane Australia	9/13/2023 9:25:00 PM	Australia
14	987 Cedar Lane, San Francisco, USA	9/27/2023 11:11:00 PM	USA
15	321 Oak Drive, Ottawa, Canada	8/31/2023 11:23:00 AM	Canada
16	654 Birch Avenue, Edinburgh, UK	8/23/2023 10:17:00 AM	UK
17	234 Maple Road, Perth, Australia	9/16/2023 2:54:00 PM	Australia
18	789 Willow Street, Miami | USA	9/2/2023 7:57:00 AM	USA

You will now have a new column that extracts the country names from the addresses.

Double-click on the heading and enter **"country"** or any name you want for the ***New Column Name****, as shown in the screenshot below.*



= Table.ReorderColumns(#"Renamed Columns",{"Address", "Registratio

	Address	Registration date	Country	$	Do
1	123 Main Street, Los Angeles, USA	9/6/2023 8:42:00 AM	USA		
2	45 Elm Avenue, Toronto, Canada	9/4/2023 7:46:00 AM	Canada		
3	789 Oak Drive, London, UK	9/16/2023 11:43:00 PM	UK		
4	567 Maple Lane, Sydney, Australia	8/16/2023 12:01:00 AM	Australia		
5	321 Pine Road, New York City, USA	9/20/2023 2:14:00 PM	USA		
6	321 Pine Road, New York City, USA	9/20/2023 2:14:00 PM	USA		
7	88 Birch Street, Vancouver, Canada	10/1/2023 9:13:00 AM	Canada		
8	89 Birch Street, Vancouver, Canada	10/2/2023 9:13:00 AM	Canada		
9	234 Cedar Avenue, Manchester, UK	8/24/2023 10:38:00 PM	UK		
10	456 Willow Court, Melbourne, Australia	8/24/2023 10:38:00 PM	Australia		
11	678 Spruce Lane, Chicago, USA	8/20/2023 6:21:00 PM	USA		
12	12 Redwood Road, Montreal, Canada	10/14/2023 7:47:00 PM	Canada		
13	890 Elm Street, Dallas, USA	4/6/2023 1:50:00 PM	USA		
14	890 Elm Street, Dallas, USA	4/6/2023 1:50:00 PM	USA		
15	76 Juniper Avenue, Brisbane, Australia	9/13/2023 9:25:00 PM	Australia		
16	987 Cedar Lane, San Francisco, USA	9/27/2023 11:11:00 PM	USA		
17	321 Oak Drive, Ottawa, Canada	8/31/2023 11:23:00 AM	Canada		
18	654 Birch Avenue, Edinburgh, UK	8/23/2023 10:17:00 AM	UK		
19	234 Maple Road, Perth, Australia	9/16/2023 2:54:00 PM	Australia		
20	789 Willow Street, Miami, USA	9/2/2023 7:57:00 AM	USA		

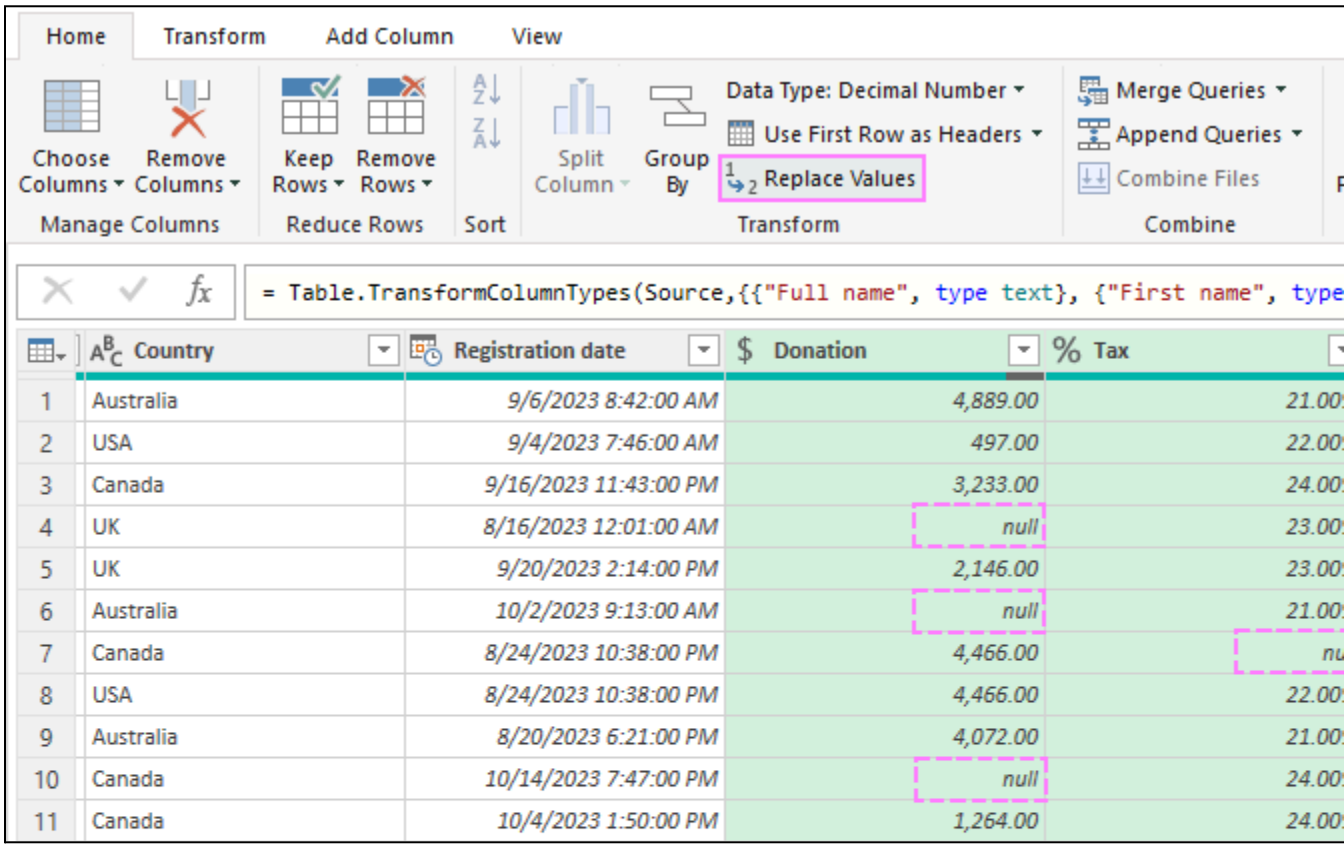
DO NOT Close & Load; we will continue this process for the next example.

Example 4. Replace missing values

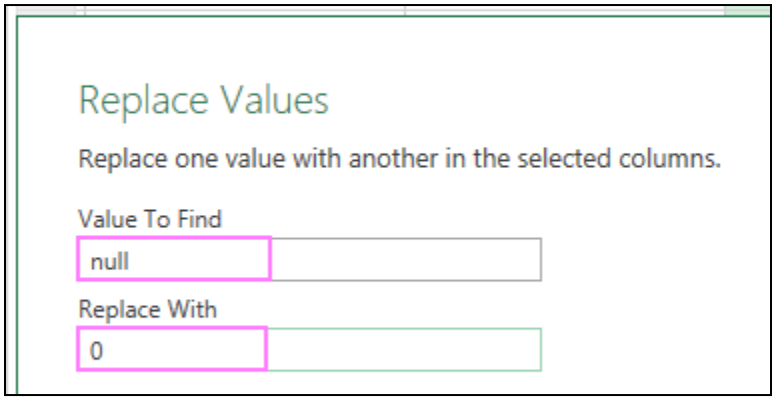
In this example, we will continue the `**example3.csv**`.

In Power Query, replacing missing values, often represented as ***null***, is a straightforward process:

1. Select the column(s) where you want to handle missing values. In our example, we can select the `**Donation**` column and `**Tax**` column.
2. On the ***Home*** tab, in the *Transform* group, click on the ***Replace Values***.



3. In the ***Replace Values*** dialog box, fill in two boxes:
    a. *Value To Find*: ***null***
    b. *Replace With*: enter the replacement value corresponding to your data type (e.g., **"0"** for numeric columns).



4. Click on the **OK** button, and Power Query will apply the replacement to all the selected columns.

	$ Donation	% Tax	A
AM	4889	21.00%	U
AM	497	22.00%	C
PM	3233	24.00%	U
AM	1376	23.00%	A
PM	0	21.00%	U
PM	2146	21.00%	U
AM	2578	22.00%	
AM	2578	22.00%	C
PM	0	24.00%	U
PM	4466	23.00%	A
PM	4072	21.00%	U
PM	3753	22.00%	C
PM	1500	24.00%	
PM	1500	21.00%	U
PM	0	23.00%	A
PM	1999	21.00%	U
AM	2236	22.00%	C

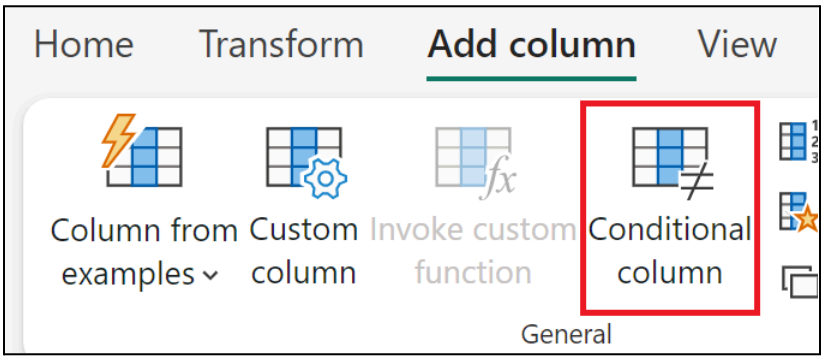
DO NOT Close & Load; we will continue this process for the next example.

Example 5. Add conditional column

In this example, we will continue the `**example3.csv**`.

To add a new column based on a set of conditions that use existing columns, make use of the ***Add Conditional Column*** feature. For example, to add a column that assigns a donor level based on the donation amount, this is what you need to do:

1. Select the **Donation** column in your dataset.
2. On the ***Add Column*** tab, click on ***Conditional Column*.**



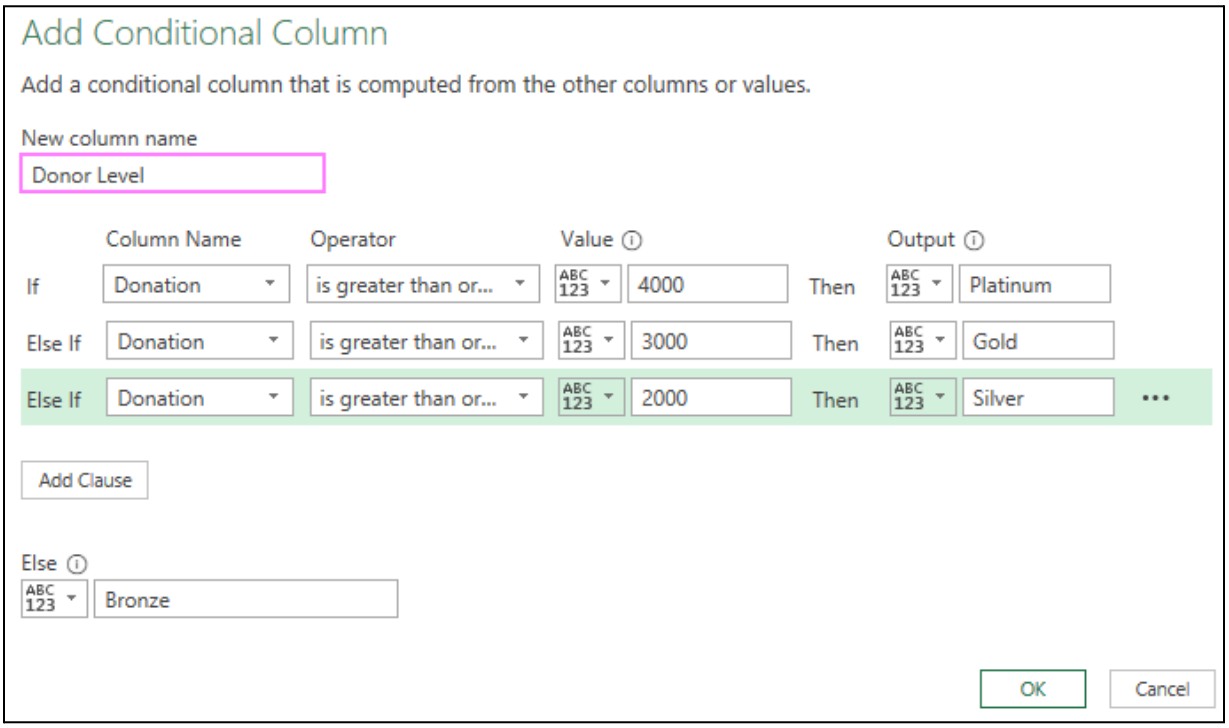
3. In the dialog box that opens, type ***Donor Level*** as the name of the new column. Then, specify the following rules, as shown in the screenshot below.

   `| If → Donation | is greater than or equal to | 4000 | Then | Platinum`

   `|Else If → Donation | is greater than or equal to | 3000 | Then | Gold`

   `|Else If Donation | is greater than or equal to | 2000 | Then | Silver`

   `| Otherwise → Bronze |`



4. Click on the **OK** button to create the new column.

This feature is similar to writing a nested `IF statement` in Excel, but it is a lot easier and more convenient to use.

By default, the new conditional column will appear at the end of your dataset, and you can drag it to any position you want.

```
= Table.AddColumn(#"Renamed Columns", "Donor Level ", each if [Donation] >= 4000 then "Platinum " else
```

Address	Registration date	Donation	Donor Level
789 Juniper Street, NSW 2000, Australia	9/6/2023 8:42:00 AM	4,889.00	Platinum
789 Willow Street, CA 90210, USA	9/4/2023 7:46:00 AM	497.00	Bronze
89 Birch Street, M5V 2T6, Canada	9/16/2023 11:43:00 PM	3,233.00	Gold
123 Spruce Avenue, SW1A 1AA, UK	8/16/2023 12:01:00 AM	1,376.00	Bronze
789 Oak Drive, SW1A 1AA, UK	9/20/2023 2:14:00 PM	2,146.00	Silver
567 Maple Lane, 2000, Australia	10/2/2023 9:13:00 AM	2,578.00	Silver
890 Birch Road, ON M5V 2T6, Canada	8/24/2023 10:38:00 PM	4,466.00	Platinum
234 Cedar Road, IA 90210, USA	8/24/2023 10:38:00 PM	4,466.00	Platinum
76 Juniper Avenue, 2000, Australia	8/20/2023 6:21:00 PM	4,072.00	Platinum
12 Redwood Road, QC M5V 2T6, Canada	10/14/2023 7:47:00 PM	3,753.00	Gold
45 Elm Avenue, M5V 2T6, Canada	10/4/2023 1:50:00 PM	1,264.00	Bronze
45 Elm Avenue, M5V 2T6, Canada	10/4/2023 1:50:00 PM	1,264.00	Bronze
567 Pine Court, QC M5V 2T6, Canada	9/13/2023 9:25:00 PM	2,666.00	Silver
234 Maple Road, SA 2000, Australia	9/27/2023 11:11:00 PM	1,999.00	Bronze

MY SCREENSHOT

```
= Table.AddColumn(#"Replaced Value", "Donor Level", each if [Donation] >= 4000 then "Platinum" else if [Donation] >= 3000 then "Gold" else if [Donation] >= 2000 then "Silver" else "Bronze")
```

	Address	Registration date	Donation	Tax	Text After Delimiter	Donor Level
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%
	26 distinct, 23 unique	26 distinct, 23 unique	21 distinct, 14 unique	4 distinct, 0 unique	4 distinct, 0 unique	
1	123 Main Street, Los Angeles, USA	9/6/2023 8:42:00 AM	4889	21.00%	USA	Platinum
2	45 Elm Avenue, Toronto, Canada	9/4/2023 7:46:00 AM	497	22.00%	Canada	Bronze
3	789 Oak Drive, London, UK	9/16/2023 11:43:00 PM	3233	24.00%	UK	Gold
4	567 Maple Lane, Sydney, Australia	8/16/2023 12:01:00 AM	1376	23.00%	Australia	Bronze
5	321 Pine Road, New York City, USA	9/20/2023 2:14:00 PM	0	21.00%	USA	Bronze
6	321 Pine Road, New York City, USA	9/20/2023 2:14:00 PM	2146	21.00%	USA	Silver
7	88 Birch Street, Vancouver, Canada	10/1/2023 9:13:00 AM	2578	22.00%	Canada	Silver
8	89 Birch Street, Vancouver, Canada	10/2/2023 9:13:00 AM	2578	22.00%	Canada	Silver
9	234 Cedar Avenue, Manchester, UK	8/24/2023 10:38:00 PM	0	24.00%	UK	Bronze
10	456 Willow Court, Melbourne, Australia	8/24/2023 10:38:00 PM	4466	23.00%	Australia	Platinum
11	678 Spruce Lane, Chicago, USA	8/20/2023 6:21:00 PM	4072	21.00%	USA	Platinum
12	12 Redwood Road, Montreal, Canada	10/14/2023 7:47:00 PM	3753	22.00%	Canada	Gold
13	890 Elm Street, Dallas, USA	4/6/2023 1:50:00 PM	1500	21.00%	USA	Bronze
14	890 Elm Street, Dallas, USA	4/6/2023 1:50:00 PM	1500	21.00%	USA	Bronze
15	76 Juniper Avenue, Brisbane, Australia	9/13/2023 9:25:00 PM	0	23.00%	Australia	Bronze
16	987 Cedar Lane, San Francisco, USA	9/27/2023 11:11:00 PM	1999	21.00%	USA	Bronze
17	321 Oak Drive, Ottawa, Canada	8/31/2023 11:23:00 AM	2236	22.00%	Canada	Silver
18	654 Birch Avenue, Edinburgh, UK	8/23/2023 10:17:00 AM	1631	24.00%	UK	Bronze
19	234 Maple Road, Perth, Australia	9/16/2023 2:54:00 PM	2679	23.00%	Australia	Silver
20	789 Willow Street, Miami, USA	9/2/2023 7:57:00 AM	3923	21.00%	USA	Gold
21	567 Pine Court, Calgary, Canada	10/26/2023 4:42:00 PM	1251	22.00%	Canada	Bronze
22	567 Pine Court, Calgary, Canada	9/16/2023 6:09:00 PM	2207	22.00%	Canada	Silver
23	345 Oak Lane, Liverpool, UK	10/6/2023 9:40:00 AM	4436	24.00%	UK	Platinum
24	12 Elm Drive, Adelaide, Australia	9/23/2023 12:03:00 PM	3428	23.00%	Australia	Gold
25	456 Redwood Lane, Houston, USA	9/25/2023 5:38:00 AM	3036	21.00%	USA	Gold
26	890 Birch Road, Toronto, Canada	10/18/2023 9:28:00 AM	1251	22.00%	Canada	Bronze
27	123 Spruce Avenue, Glasgow, UK	9/29/2023 8:13:00 PM	4072	24.00%	UK	Platinum
28	789 Juniper Street, Sydney, Australia	9/3/2023 2:55:00 PM	2146	23.00%	Australia	Silver
29	234 Cedar Road, San Diego, USA	8/21/2023 3:59:00 PM	2679	21.00%	USA	Silver

A conditional column essentially uses an IF function to populate the values

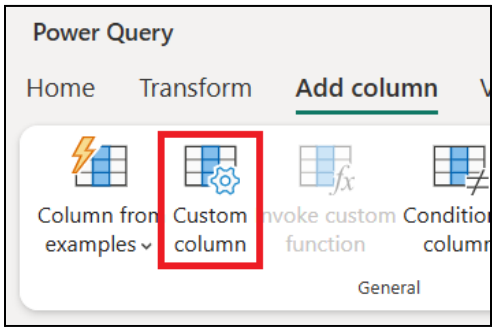
Example 6. Write formulas in Power Query

In this example, we will continue the `**example3.csv**`.

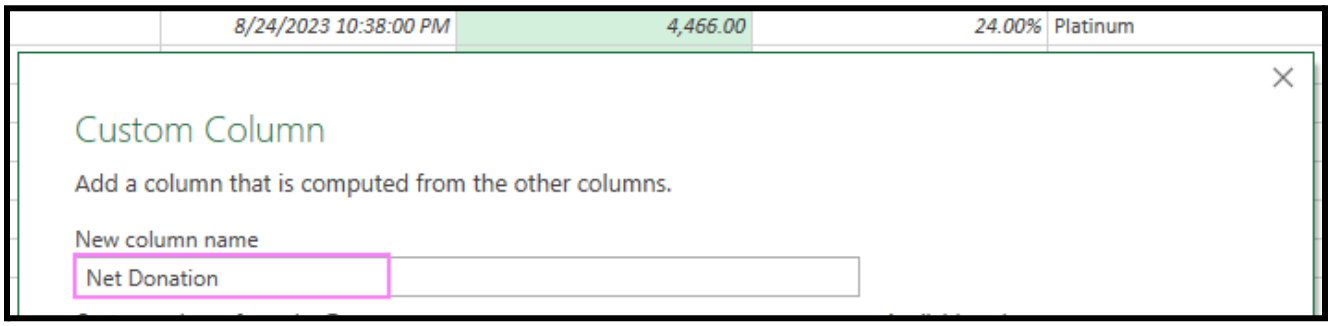
Similar to using functions and formulas in Excel worksheets, PQE also has its set of functions to perform various calculations. These formulas are written in the Power Query formula language, also known as M. Now, here's the great news: you do not need to learn the intricacies of M language syntax to build your custom formulas. Unlike traditional coding, Power Query makes it wonderfully simple and user-friendly using a visual interface.

Let's take an example. Imagine you want to create a column that calculates the net donation amount after tax deduction based on the values in the Donation and Tax Rate columns. The steps are:

1. Go to the ***Add Column*** tab and click on the ***Custom Column***.



2. In the ***Custom Column*** dialog box, type**" Net Donation"** or any name you want for the *New Column Name, As*



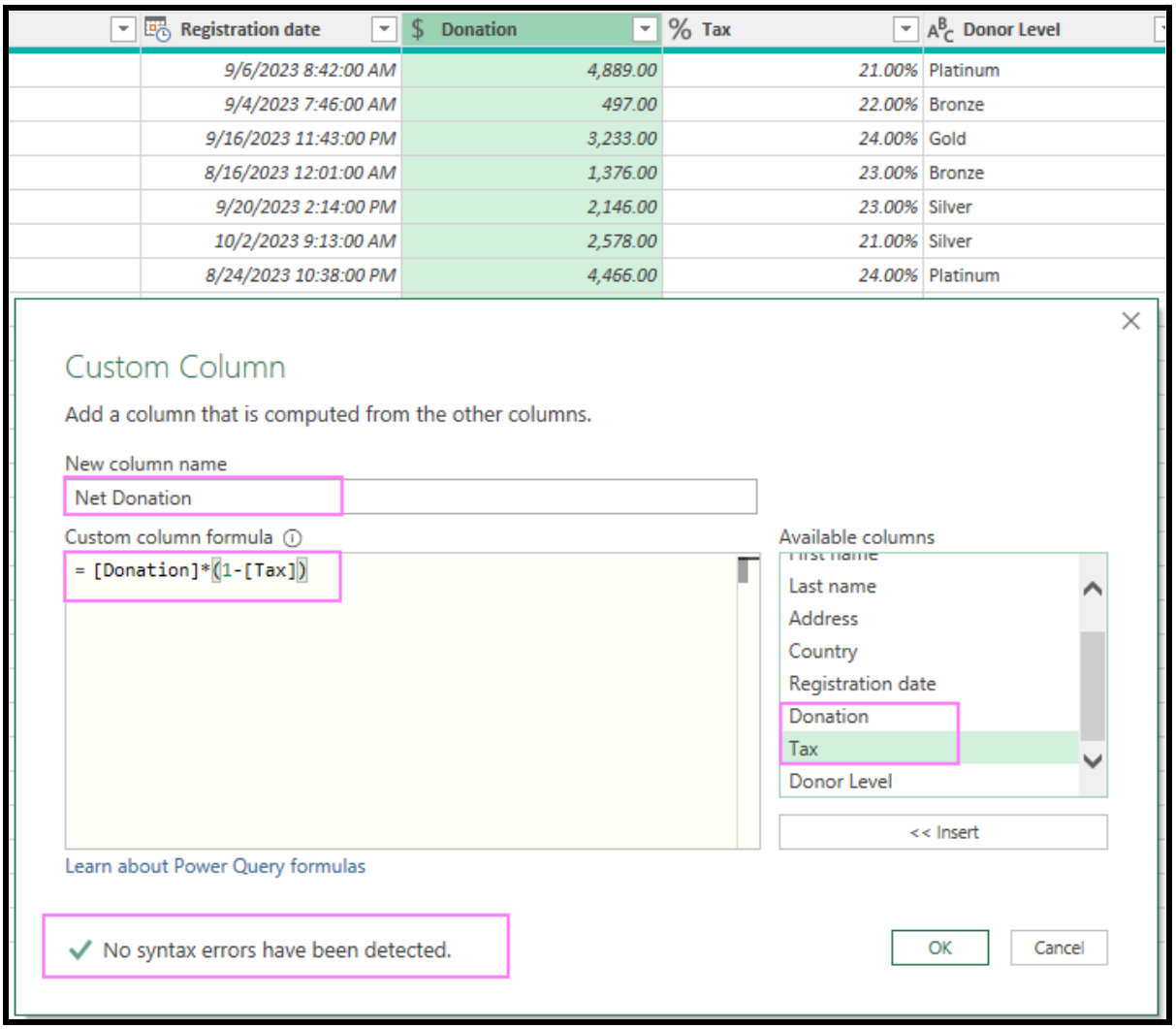
3. The ***Custom Column*** *Formula* box is where you write the formula itself. So, enter the following formula in it:

```
=[Donation]*(1-[Tax])
```

To refer to a specific column in the formula, select its name under *Available Columns* on the right side of the dialog and click *Insert*, or just double-click the column name.

4. As you enter a formula, Power Query checks it, and if no errors are found, it displays the message “No syntax errors have been detected." If there’s an error in your formula, it will show an error message indicating the error type.

5. Click on the **OK** button, and the formula will be applied to all rows in the new column.



That's it! You have successfully created a custom column with a formula in Power Query. The formula bar shows its syntax.

```
= Table.AddColumn(#"Changed Type", "Net Donation", each [Donation]*(1-[Tax]))
```

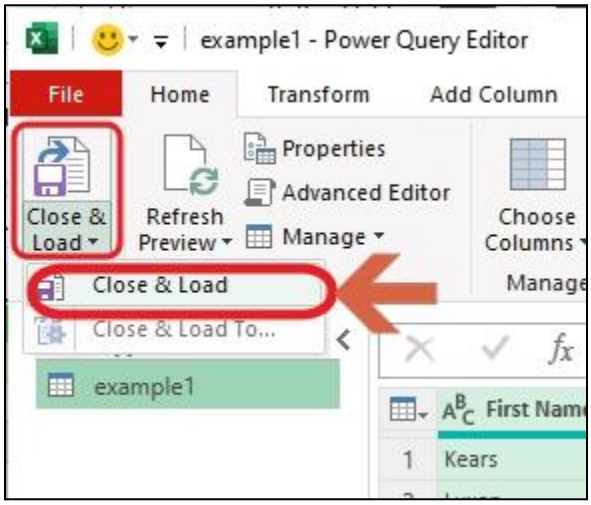
	Registration date	$ Donation	% Tax	$ Net Donation
1	9/6/2023 8:42:00 AM	4,889.00	21.00%	3,862.31
2	9/4/2023 7:46:00 AM	497.00	22.00%	387.66
3	9/16/2023 11:43:00 PM	3,233.00	24.00%	2,457.08
4	8/16/2023 12:01:00 AM	1,376.00	23.00%	1,059.52
5	9/20/2023 2:14:00 PM	2,146.00	23.00%	1,652.42
6	10/2/2023 9:13:00 AM	2,578.00	21.00%	2,036.62
7	8/24/2023 10:38:00 PM	4,466.00	24.00%	3,394.16
8	8/24/2023 10:38:00 PM	4,466.00	22.00%	3,483.48
9	8/20/2023 6:21:00 PM	4,072.00	21.00%	3,216.88
10	10/14/2023 7:47:00 PM	3,753.00	24.00%	2,852.28
11	10/4/2023 1:50:00 PM	1,264.00	24.00%	960.64
12	10/4/2023 1:50:00 PM	1,264.00	24.00%	960.64
13	9/13/2023 9:25:00 PM	2,666.00	24.00%	2,026.16
14	9/27/2023 11:11:00 PM	1,999.00	21.00%	1,579.21
15	8/31/2023 11:23:00 AM	2,236.00	21.00%	1,766.44
16	8/23/2023 10:17:00 AM	1,631.00	23.00%	1,255.87
17	9/16/2023 2:54:00 PM	2,679.00	22.00%	2,089.62
18	9/2/2023 7:57:00 AM	3,923.00	21.00%	3,099.17
19	10/26/2023 4:42:00 PM	1,251.00	22.00%	975.78
20	10/6/2023 9:40:00 AM	4,436.00	22.00%	3,460.08
21	9/23/2023 12:03:00 PM	3,428.00	24.00%	2,605.28
22	9/25/2023 5:38:00 AM	3,036.00	23.00%	2,337.72

MY SCREENSHOT

Text After Delimiter	Donor Level	Net Donation
Valid 100% Error 0% Empty 0% 4 distinct, 0 unique	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
USA	Platinum	3862.31
Canada	Bronze	387.66
UK	Gold	2457.08
Australia	Bronze	1059.52
USA	Bronze	0
USA	Silver	1695.34
Canada	Silver	2010.84
Canada	Silver	2010.84
UK	Bronze	0
Australia	Platinum	3438.82
USA	Platinum	3216.88

Here the values displayed came from a custom formula to deduct the tax percentage

18. We are now ready to load it into our Excel worksheet by selecting **Close & Load → Close & load, as shown in the screenshot below.**



MY SCREENSHOT

Address	Registration date	Donation	Tax	Text After Delimiter	Donor Level	Net Donation
123 Main Street, Los Angeles, USA	9/6/2023 8:42	$4,889	0.21	USA	Platinum	$3,862
45 Elm Avenue, Toronto, Canada	9/4/2023 7:46	$497	0.22	Canada	Bronze	$388
789 Oak Drive, London, UK	9/16/2023 23:43	$3,233	0.24	UK	Gold	$2,457
567 Maple Lane, Sydney, Australia	8/16/2023 0:01	$1,376	0.23	Australia	Bronze	$1,060
321 Pine Road, New York City, USA	9/20/2023 14:14	$0	0.21	USA	Bronze	$0
321 Pine Road, New York City, USA	9/20/2023 14:14	$2,146	0.21	USA	Silver	$1,695
88 Birch Street, Vancouver, Canada	10/1/2023 9:13	$2,578	0.22	Canada	Silver	$2,011
89 Birch Street, Vancouver, Canada	10/2/2023 9:13	$2,578	0.22	Canada	Silver	$2,011
234 Cedar Avenue, Manchester, UK	8/24/2023 22:38	$0	0.24	UK	Bronze	$0
456 Willow Court, Melbourne, Australia	8/24/2023 22:38	$4,466	0.23	Australia	Platinum	$3,439
678 Spruce Lane, Chicago, USA	8/20/2023 18:21	$4,072	0.21	USA	Platinum	$3,217
12 Redwood Road, Montreal, Canada	10/14/2023 19:47	$3,753	0.22	Canada	Gold	$2,927
890 Elm Street, Dallas, USA	4/6/2023 13:50	$1,500	0.21	USA	Bronze	$1,185
890 Elm Street, Dallas, USA	4/6/2023 13:50	$1,500	0.21	USA	Bronze	$1,185
76 Juniper Avenue, Brisbane, Australia	9/13/2023 21:25	$0	0.23	Australia	Bronze	$0
987 Cedar Lane, San Francisco, USA	9/27/2023 23:11	$1,999	0.21	USA	Bronze	$1,579
321 Oak Drive, Ottawa, Canada	8/31/2023 11:23	$2,236	0.22	Canada	Silver	$1,744
654 Birch Avenue, Edinburgh, UK	8/23/2023 10:17	$1,631	0.24	UK	Bronze	$1,240
234 Maple Road, Perth, Australia	9/16/2023 14:54	$2,679	0.23	Australia	Silver	$2,063
789 Willow Street, Miami, USA	9/2/2023 7:57	$3,923	0.21	USA	Gold	$3,099
567 Pine Court, Calgary, Canada	10/26/2023 16:42	$1,251	0.22	Canada	Bronze	$976
567 Pine Court, Calgary, Canada	9/16/2023 18:09	$2,207	0.22	Canada	Silver	$1,721
345 Oak Lane, Liverpool, UK	10/6/2023 9:40	$4,436	0.24	UK	Platinum	$3,371
12 Elm Drive, Adelaide, Australia	9/23/2023 12:03	$3,428	0.23	Australia	Gold	$2,640
456 Redwood Lane, Houston, USA	9/25/2023 5:38	$3,036	0.21	USA	Gold	$2,398
890 Birch Road, Toronto, Canada	10/18/2023 9:28	$1,251	0.22	Canada	Bronze	$976
123 Spruce Avenue, Glasgow, UK	9/29/2023 20:13	$4,072	0.24	UK	Platinum	$3,095
789 Juniper Street, Sydney, Australia	9/3/2023 14:55	$2,146	0.23	Australia	Silver	$1,652
234 Cedar Road, San Diego, USA	8/21/2023 15:59	$2,679	0.21	USA	Silver	$2,116

After Closing and Loading, I noticed that when clicking on the values, the formulas used to create them aren't visible.

Example 7: Grouping and Aggregating Data Using Power Query

In this example, we will continue the `**example3.csv**`.

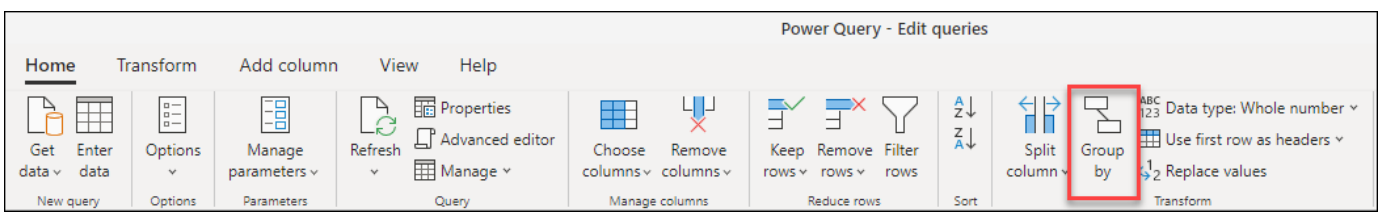
Summarizing or aggregating data by specific groups is a common task in data analysis. Power Query simplifies this process with its powerful **Group By** feature. Here's how you can use this function to aggregate data effectively.

Scenario:

You have a dataset containing donation records, including columns for **Country**, **Donor Level**, and **Donation Amount**. You want to calculate the total donation amount grouped by **Country** and **Donor Level**.

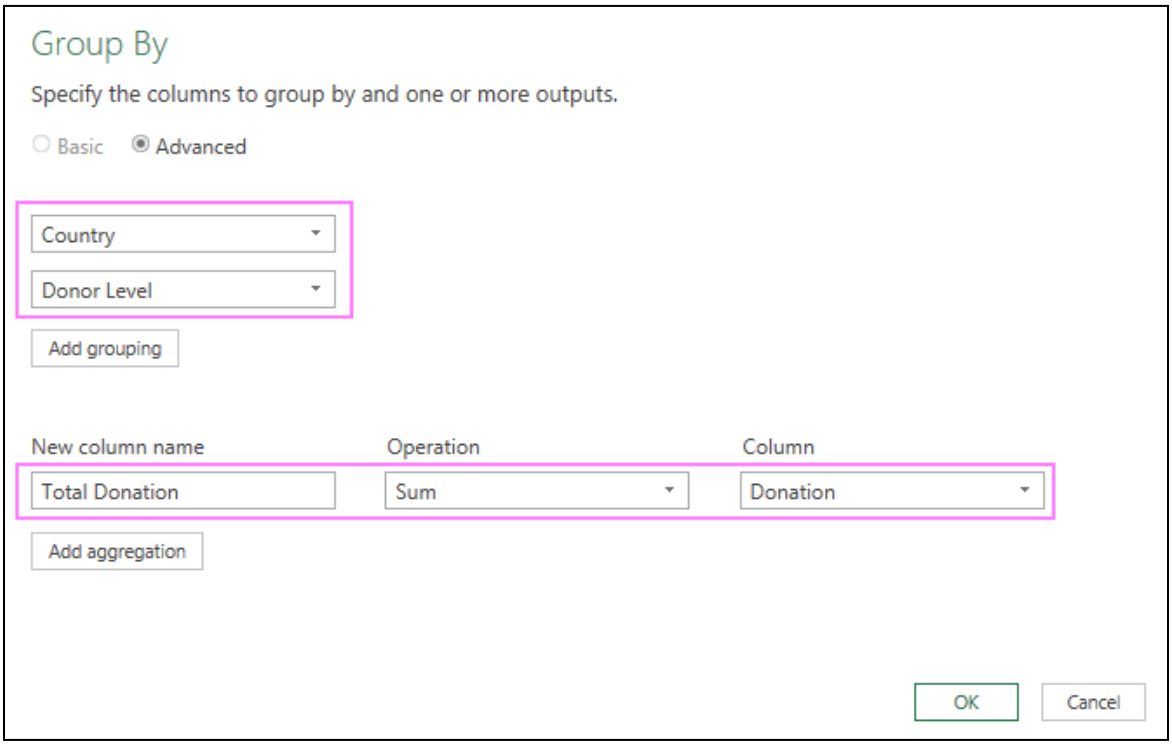
Following are a Steps to Group and Aggregate Data:

1. **Access the Group By Feature:**
   - On the **Home** tab in the Power Query Editor ribbon, click **Group By**.



2. **Configure the Group By Settings:**
   - Group By dialog box will appear. In the dialog box, select **Advanced** to group by multiple columns.
   - Perform the following steps, as shown in the screenshot below.
     - **Under Group By:** Select **Country** and **Donor Level** as the columns to group by.
     - **Under New Column Name:** Type "Total Donation" as the name of the new aggregated column.

- **Under Operation:** Select **Sum** as the aggregation function.
- **Under Column:** Choose **Donation** as the column to aggregate.



- Click on the **OK** button to apply the grouping and aggregation.

3. **Optional: Sort the Aggregated Data:**
   - If needed, sort the table to organize the data:
     - Click the filter arrow next to a column header (e.g., **Total Donation**).
     - Choose to sort in ascending or descending order.

Outcome:

After completing these steps:

- A new table will be created displaying the grouped and aggregated data.
- Each row in the table will represent a unique combination of **Country** and **Donor Level**, along with the corresponding **Total Donation** value, as shown in the screenshot below.

	Country	Donor Level	Total Donation
1	Australia	Silver	4825
2	Australia	Gold	3428
3	Australia	Platinum	4466
4	Australia	Bronze	1376
5	Canada	Silver	9599
6	Canada	Gold	3753
7	Canada	Bronze	2999
8	UK	Platinum	8508
9	UK	Gold	3233
10	UK	Bronze	1631
11	USA	Gold	6959
12	USA	Silver	4825
13	USA	Bronze	4999
14	USA	Platinum	8961

MY SCREENSHOT

Country	Donor Level	Total Donation
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
4 distinct, 0 unique		13 distinct, 12 unique
Canada	Silver	9,599.00
USA	Platinum	8,961.00
UK	Platinum	8,508.00
USA	Gold	6,959.00
USA	Bronze	4,999.00
USA	Silver	4,825.00
Australia	Silver	4,825.00
Australia	Platinum	4,466.00
Canada	Gold	3,753.00
Australia	Gold	3,428.00
UK	Gold	3,233.00
Canada	Bronze	2,999.00
UK	Bronze	1,631.00
Australia	Bronze	1,376.00

Reference: https://learn.microsoft.com/en-us/power-query/group-by

Exercise 8: Removing Duplicate Data Using Power Query

In this example, we will use the `**example4.csv**` file.

Removing duplicate data is a crucial step in ensuring the accuracy and integrity of your dataset. Duplicate records can lead to significant errors, such as overestimating reported sales, which can have serious consequences for your business. Power Query simplifies this process by providing an intuitive feature to remove duplicates directly within its interface. Here's how you can effectively handle duplicate data using Power Query:

Scenario:

In our dataset, the record for **“Stevie Bridge”** appears twice under the **[Sales]** column for **"Asia" with a value of 4378**. However, one record spells **"Bridge"** with an uppercase **"B"** and the other with a lowercase "b."

- **Key Point:** Power Query is case-sensitive when analyzing data, meaning it treats "Bridge" and "bridge" as distinct. However, the Remove Duplicates feature is case-insensitive, so it treats these as duplicates once the casing is consistent.

As shown in the screenshot below.

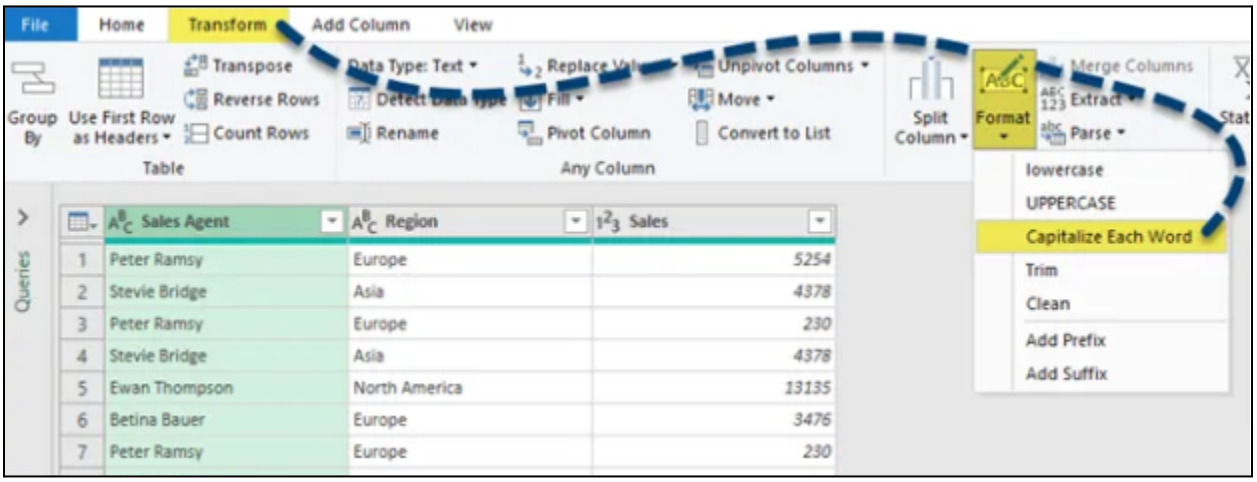
Sales Agent	Region	Sales
Peter Ramsy	Europe	5254
Stevie Bridge	Asia	4378
Peter Ramsy	Europe	230
Ewan Thompson	North America	13135
Betina Bauer	Europe	3476
Peter Ramsy	Asia	1490
Stevie Bridge	Europe	497
Betina Bauer	Asia	9267
Stevie Bridge	North America	927
Stevie bridge	Asia	4378
Betina Bauer	Australia	463
Peter Ramsy	South America	2317
Stevie Bridge	Australia	5361
Ewan Thompson	Asia	4468
Ewan Thompson	Europe	4468

In this example, we will use the **`example 4`** worksheet.

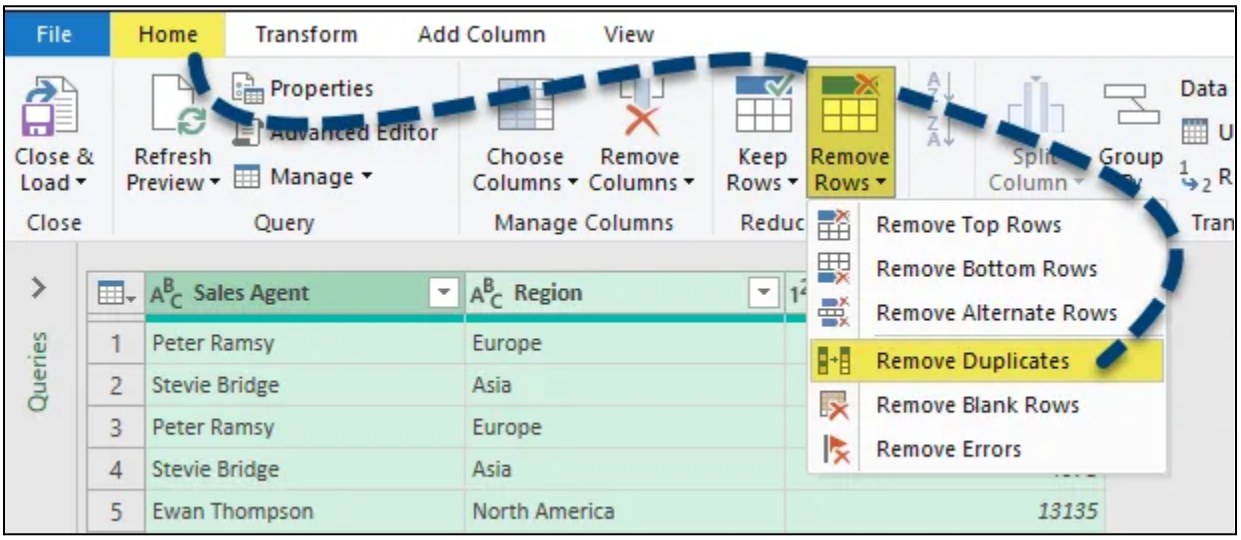
1. Go to **Data** tab → **Get Data** → **From File** → click on **From Text/CSV**.
2. In the *Import Data* dialog box, select the desired **CSV** file to import it; in our case, select **example3.csv**. Click the **Import** button.
3. Click on the **Transform Dat**a button.

Following are the steps to remove duplicates:

1. **Ensure Consistent Casing:**
   - Before removing duplicates, standardize the case for the column where duplicates exist.
   - Use Power Query's transformation options to capitalize the first letter of each word:
     - Go to the **Transform** tab in the Power Query Editor ribbon.
     - Select the column containing the names (e.g., [Name]).
     - Click on **Format** → **Capitalize Each Word** to make the casing uniform.

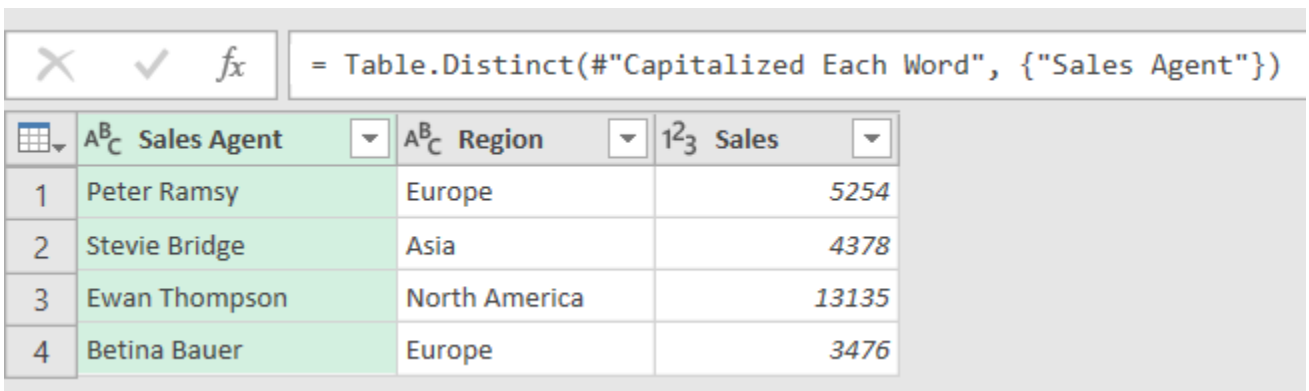


2. **Remove Duplicates:**
   - After standardizing the casing, you can proceed to remove duplicates:
     - Select **Region** column → Navigate to the **Home** tab → Click the **Remove Rows** button and select **Remove Duplicates** from the drop-down menu.



- Alternatively, **right-click** on the column header and select Remove Duplicates from the context menu.

3. **Review and confirm:**
   - Check the results to ensure duplicates have been removed. Only unique records will remain in your dataset.



MY SCREENSHOT

	Sales Agent	Region	Sales
	Valid 100% Error 0% Empty 0% 4 distinct, 4 unique	Valid 100% Error 0% Empty 0% 3 distinct, 2 unique	Valid 100% Error 0% Empty 0% 4 distinct, 4 unique
1	Peter Ramsy	Europe	*5254*
2	Stevie Bridge	Asia	*4378*
3	Ewan Thompson	North America	*13135*
4	Betina Bauer	Europe	*3476*

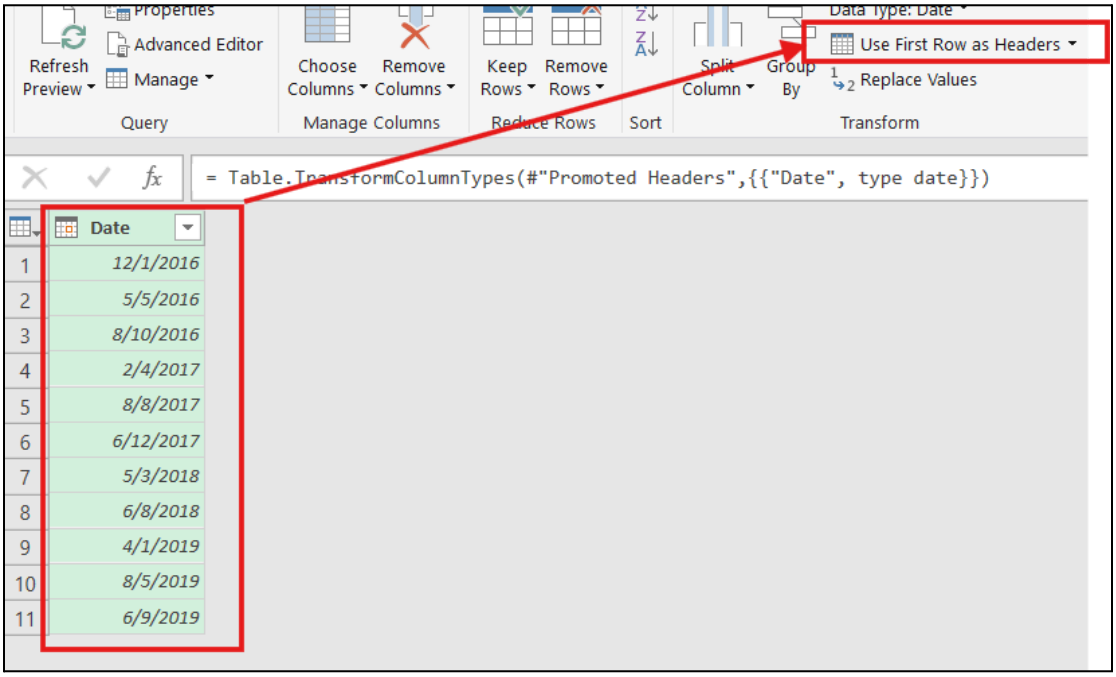
In order to remove the duplicates, it was important to standardize the text cases first as PQ is case sensitive.

Example 9: Manipulate with Date and Time

In this example, we will manipulate the date. First, we will import a csv file. In this example, we will use the **example5.csv** file.

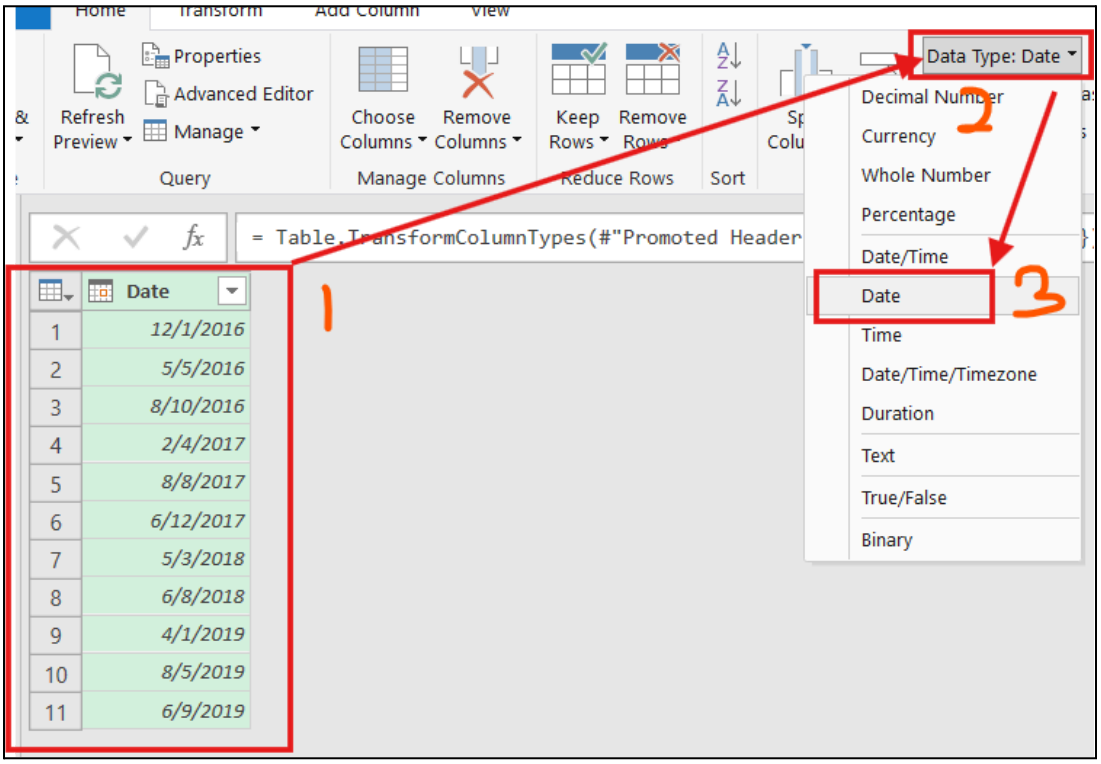
First, we will make the first row a header, as shown below.

1. Select **Date** column → Select **`Use First Row as Header`**



It is best practice that whenever we work on a `**Date**` column, we must convert that column to the **'Data Type:Date'**, as shown in the screenshot below.

2. Select **Date** column → On the ***Home*** tab, Click on **Data type** option → Select `**Date**` option

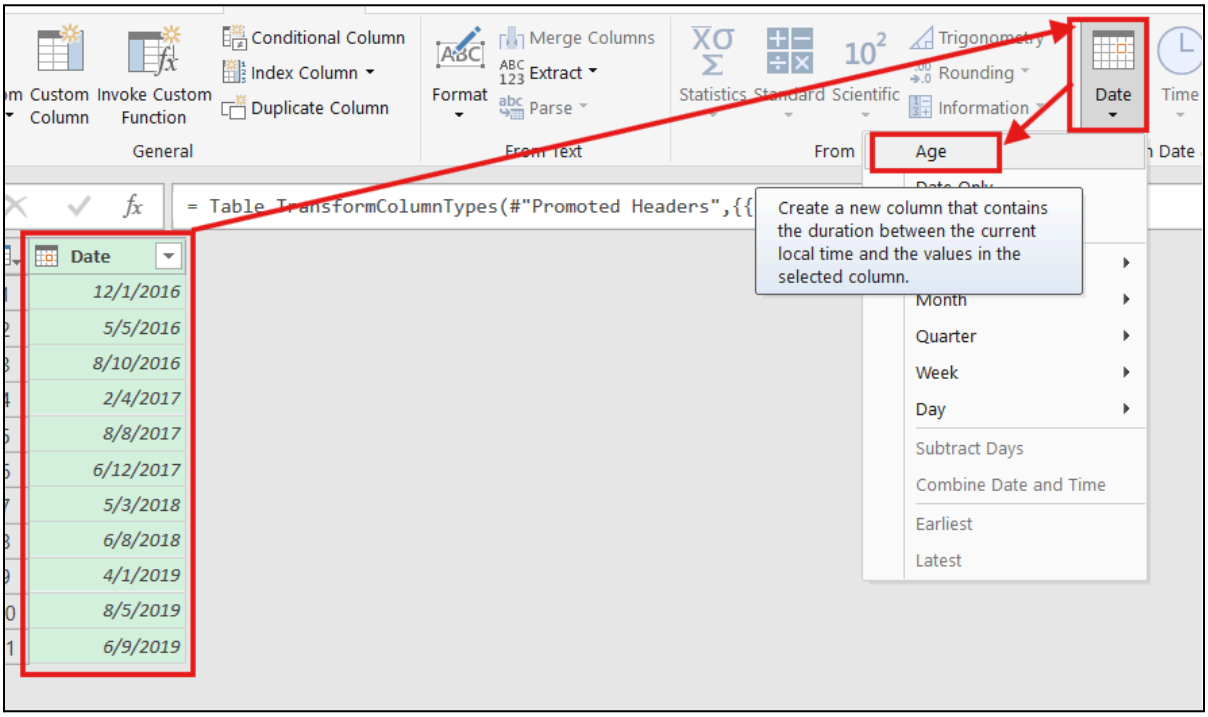


9.1. Calculate age

Calculating someone's or something's age is a common need in any type of analysis, and Power Query has this calculation built in.

The number of days between the specified date and the current date is determined when you choose the **Age** option and choose a date column. Next, you must swap out the current date for the one in the second date column.

1. Select the **Date** column. Navigate to the **Add column** tab on the top of the ribbon and then **go to the Date and select Age.**



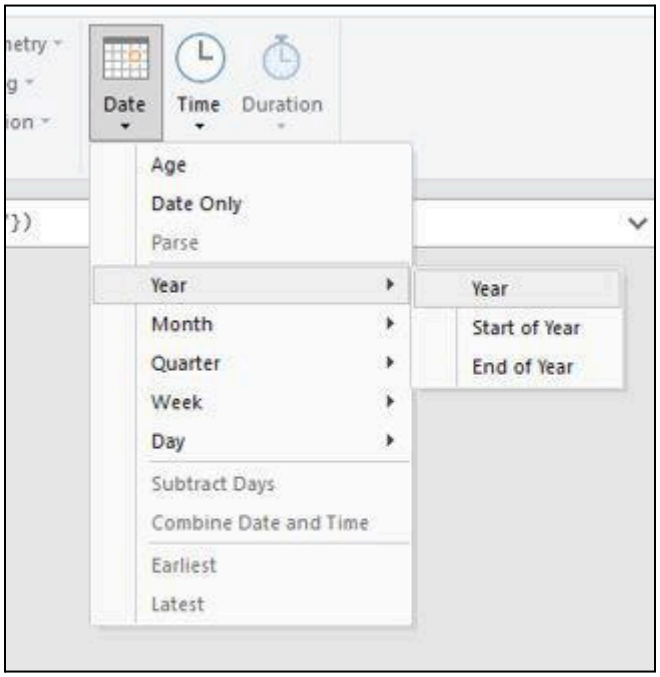
Now, as you can see with this option, **the Date value is transformed into the Age** based time format: **Days.Hours:Minutes: Seconds or 2939.00:00:00.** As shown in the screenshot below.

	Date	Age
1	12/1/2016	2939.00:00:00
2	5/5/2016	3149.00:00:00
3	8/10/2016	3052.00:00:00
4	2/4/2017	2874.00:00:00
5	8/8/2017	2689.00:00:00
6	6/12/2017	2746.00:00:00
7	5/3/2018	2421.00:00:00
8	6/8/2018	2385.00:00:00
9	4/1/2019	2088.00:00:00
10	8/5/2019	1962.00:00:00
11	6/9/2019	2019.00:00:00

9.2. Split Year from Date

When you choose the **Year** option while selecting a date column, just the year is displayed and the date and time are not displayed. It adds a new column with the name of the year and displays only the year. There are several choices available: the **`Year`**, the **`Start of the Year`**, and the **`End of the Year.`**

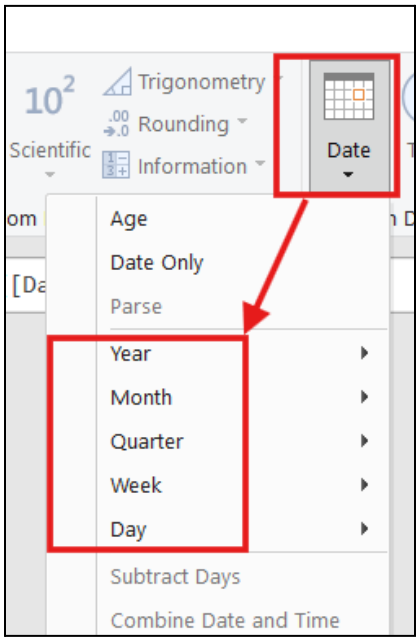
1. Select the date column. Navigate to the add column tab on the top of the ribbon and then go to the Date and select Year.



2. Now,as you can see with this option, the **Date** value is transformed into the **Year**.

	Date	Age	Year
1	12/1/2016	2939.00:00:00	2016
2	5/5/2016	3149.00:00:00	2016
3	8/10/2016	3052.00:00:00	2016
4	2/4/2017	2874.00:00:00	2017
5	8/8/2017	2689.00:00:00	2017
6	6/12/2017	2746.00:00:00	2017
7	5/3/2018	2421.00:00:00	2018
8	6/8/2018	2385.00:00:00	2018
9	4/1/2019	2088.00:00:00	2019
10	8/5/2019	1962.00:00:00	2019
11	6/9/2019	2019.00:00:00	2019

Now, you can try yourself to explore other options such as **Month, Quarter, Week, Day.**



MY SCREENSHOT

	12/1/2016	Age	Year	Month	Day
	Valid 100% Error 0% Empty 0% 10 distinct, 10 unique	Valid 100% Error 0% Empty 0% 10 distinct, 10 unique	Valid 100% Error 0% Empty 0% 4 distinct, 0 unique	Valid 100% Error 0% Empty 0% 5 distinct, 2 unique	Valid 100% Error 0% Empty 0% 8 distinct, 6 unique
1	5/5/2016	3513.00:00:00	2016	5	5
2	8/10/2016	3416.00:00:00	2016	8	10
3	2/4/2017	3238.00:00:00	2017	2	4
4	8/8/2017	3053.00:00:00	2017	8	8
5	6/12/2017	3110.00:00:00	2017	6	12
6	5/3/2018	2785.00:00:00	2018	5	3
7	6/8/2018	2749.00:00:00	2018	6	8
8	4/1/2019	2452.00:00:00	2019	4	1
9	8/5/2019	2326.00:00:00	2019	8	5
10	6/9/2019	2383.00:00:00	2019	6	9

I was able to extract the years, months, and days easily from the date and created new columns for them quickly without the use of a TEXT function

Reference:

- https://learn.microsoft.com/en-us/power-query/column-from-example
- https://learn.microsoft.com/en-us/power-query/group-by
- M function reference
- M language document
- M type reference

Submission Guideline:

Include the following deliverables in your submission:

- Take a screenshot of your wworksheets.Submit your source file, including sa screenshot using the Start Assignment button in the top-right corner of the Assignment page in Canvas.